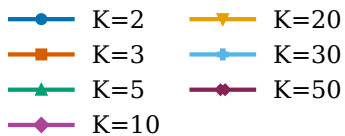
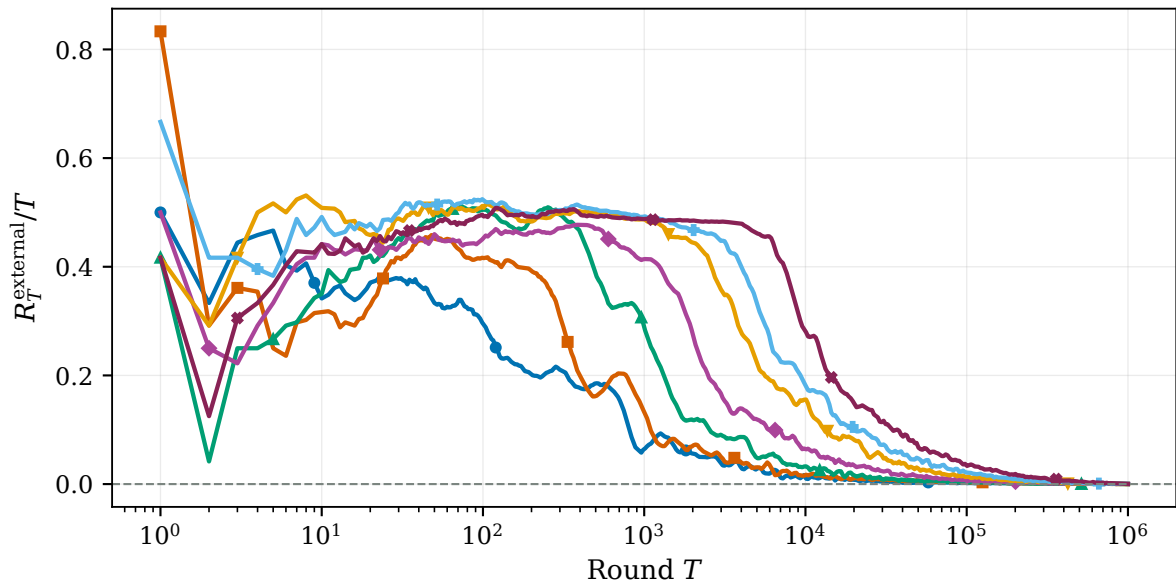


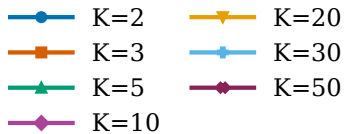
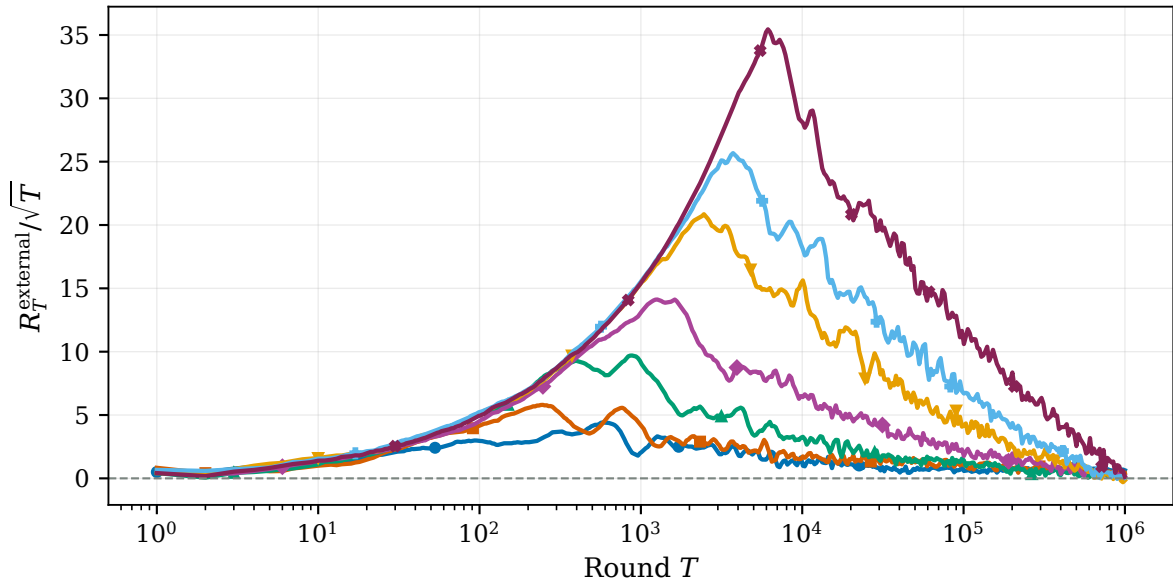
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Full information
Algorithm: BM-Hedge
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: External regret
View: R / T
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.000635083, SD = 0.000366967, SE = 0.000105934
K=3: mean = 0.000418333, SD = 0.000720775, SE = 0.00020807
K=5: mean = 1.33333e-05, SD = 0.000549218, SE = 0.000158546
K=10: mean = 0.000175083, SD = 0.000465202, SE = 0.000134292
K=20: mean = -0.000156333, SD = 0.00103618, SE = 0.000299118
K=30: mean = 4.775e-05, SD = 0.000703584, SE = 0.000203107
K=50: mean = 0.000106, SD = 0.00141069, SE = 0.000407231



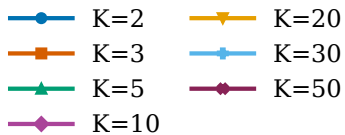
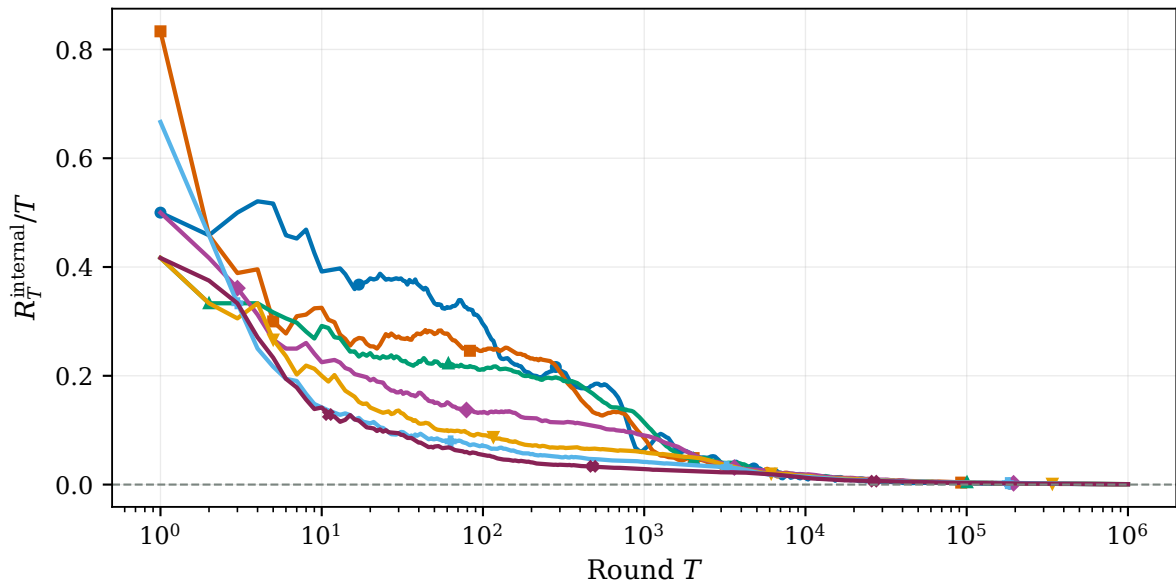
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Full information
Algorithm: BM-Hedge
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: External regret
View: $R / \sqrt{T}$
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.635083, SD = 0.366967, SE = 0.105934
K=3: mean = 0.418333, SD = 0.720775, SE = 0.20807
K=5: mean = 0.0133333, SD = 0.549218, SE = 0.158546
K=10: mean = 0.175083, SD = 0.465202, SE = 0.134292
K=20: mean = -0.156333, SD = 1.03618, SE = 0.299118
K=30: mean = 0.04775, SD = 0.703584, SE = 0.203107
K=50: mean = 0.106, SD = 1.41069, SE = 0.407231



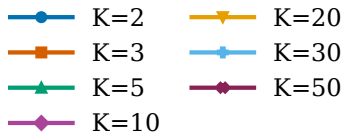
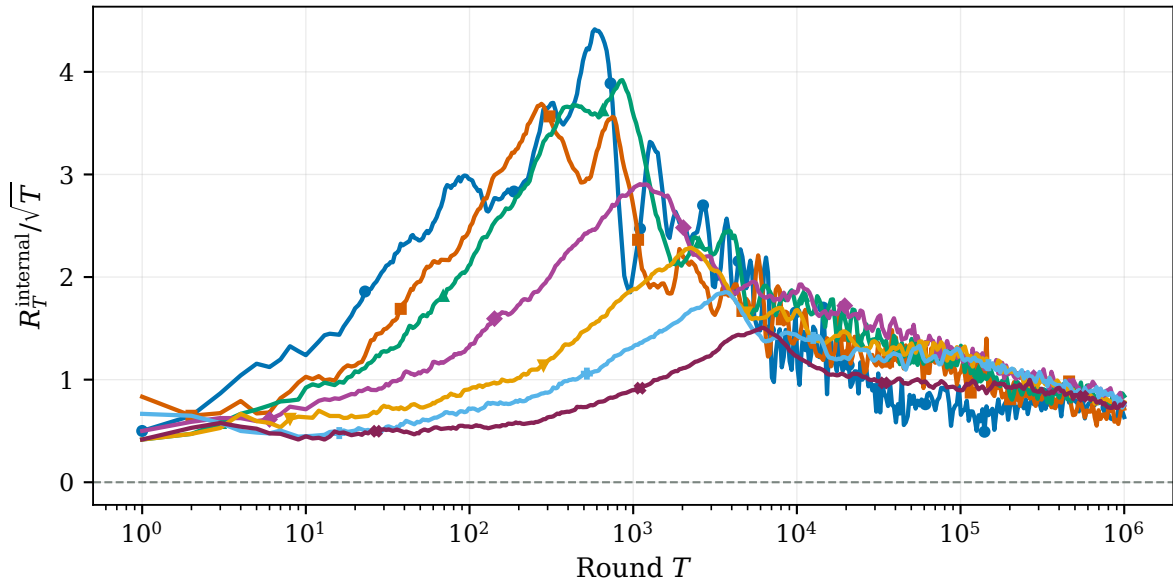
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Full information
Algorithm: BM-Hedge
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Internal regret
View: R / T
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.000635083, SD = 0.000366967, SE = 0.000105934
K=3: mean = 0.000717167, SD = 0.000431053, SE = 0.000124434
K=5: mean = 0.000839917, SD = 0.000254374, SE = 7.34313e-05
K=10: mean = 0.000749333, SD = 0.000111192, SE = 3.20984e-05
K=20: mean = 0.00078125, SD = 9.27059e-05, SE = 2.67619e-05
K=30: mean = 0.000760583, SD = 0.00012709, SE = 3.66878e-05
K=50: mean = 0.000774, SD = 9.72934e-05, SE = 2.80862e-05



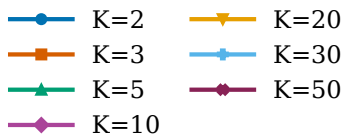
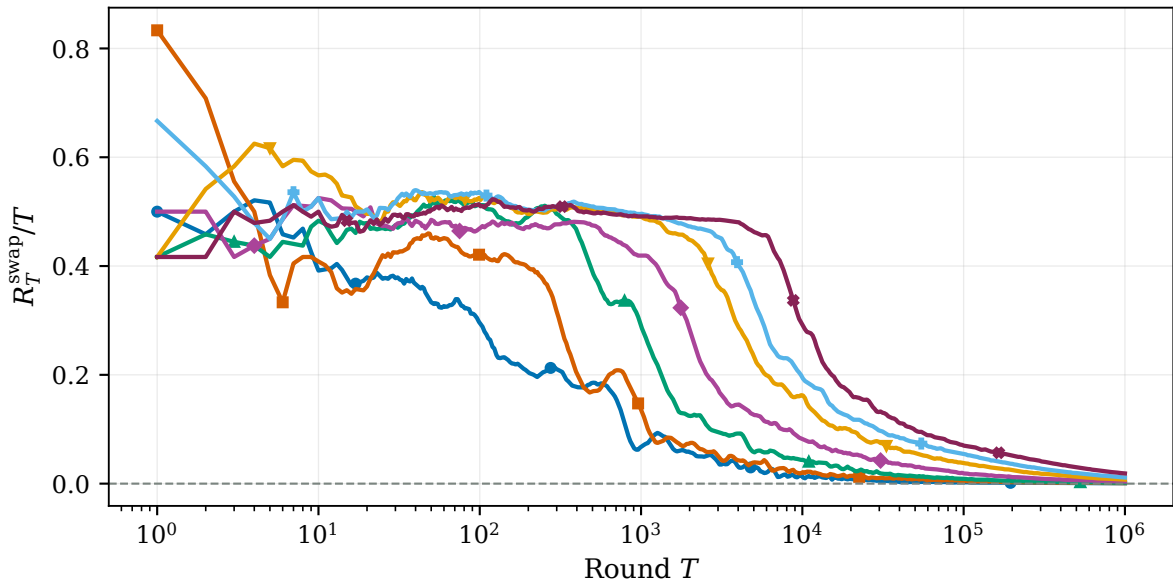
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Full information
Algorithm: BM-Hedge
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Internal regret
View: $R / \sqrt{T}$
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.635083, SD = 0.366967, SE = 0.105934
K=3: mean = 0.717167, SD = 0.431053, SE = 0.124434
K=5: mean = 0.839917, SD = 0.254374, SE = 0.0734313
K=10: mean = 0.749333, SD = 0.111192, SE = 0.0320984
K=20: mean = 0.78125, SD = 0.0927059, SE = 0.0267619
K=30: mean = 0.760583, SD = 0.12709, SE = 0.0366878
K=50: mean = 0.774, SD = 0.0972934, SE = 0.0280862



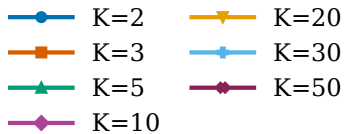
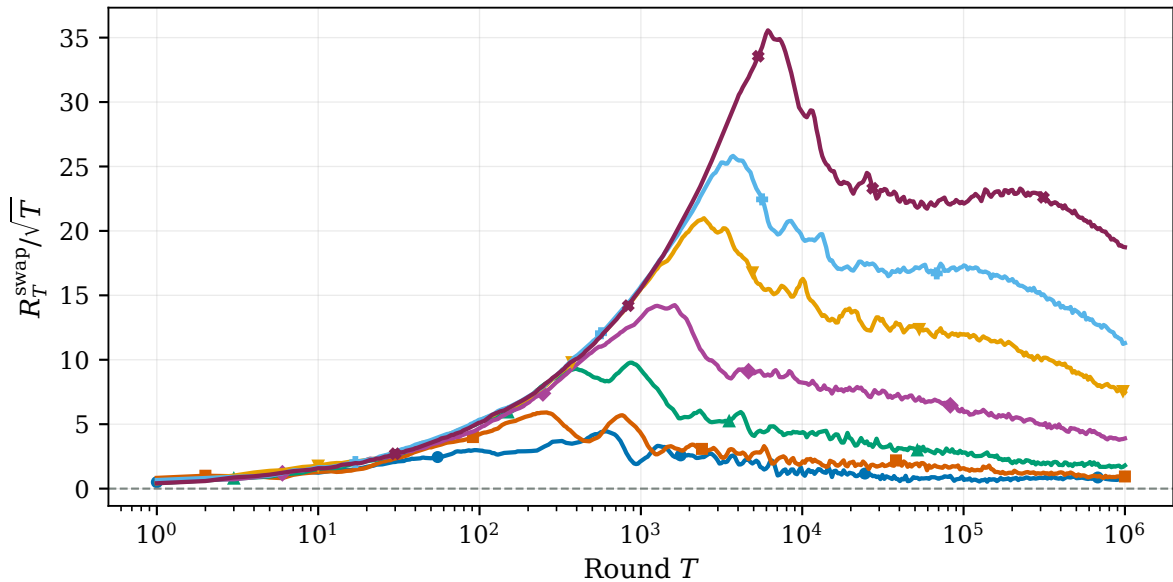
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Full information
Algorithm: BM-Hedge
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Swap regret
View: R / T
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.000727417, SD = 0.000379915, SE = 0.000109672
K=3: mean = 0.000944083, SD = 0.000591882, SE = 0.000170862
K=5: mean = 0.00179592, SD = 0.000603432, SE = 0.000174196
K=10: mean = 0.00389417, SD = 0.000468079, SE = 0.000135123
K=20: mean = 0.00746258, SD = 0.000641791, SE = 0.000185269
K=30: mean = 0.0112866, SD = 0.000830456, SE = 0.000239732
K=50: mean = 0.0187351, SD = 0.000405425, SE = 0.000117036



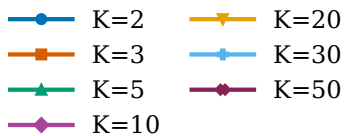
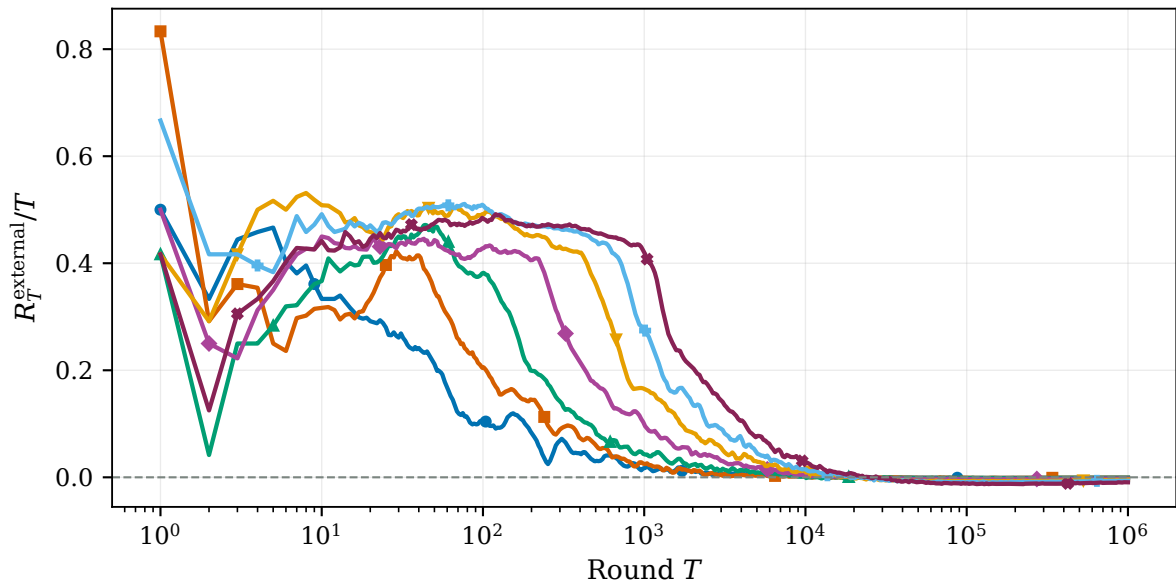
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Full information
Algorithm: BM-Hedge
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Swap regret
View: $R / \sqrt{T}$
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.727417, SD = 0.379915, SE = 0.109672
K=3: mean = 0.944083, SD = 0.591882, SE = 0.170862
K=5: mean = 1.79592, SD = 0.603432, SE = 0.174196
K=10: mean = 3.89417, SD = 0.468079, SE = 0.135123
K=20: mean = 7.46258, SD = 0.641791, SE = 0.185269
K=30: mean = 11.2866, SD = 0.830456, SE = 0.239732
K=50: mean = 18.7351, SD = 0.405425, SE = 0.117036



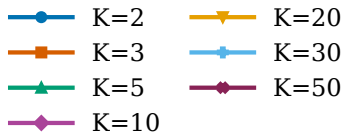
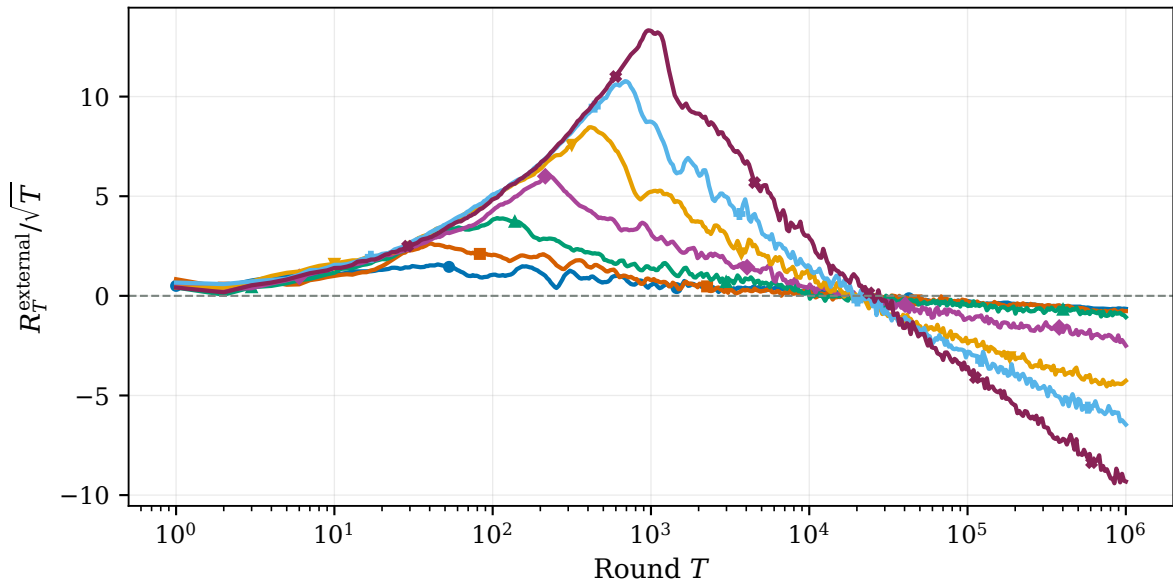
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Full information
Algorithm: BM-OptHedge
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: External regret
View: R / T
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = -0.000653833, SD = 0.000654497, SE = 0.000188937
K=3: mean = -0.000763917, SD = 0.000388655, SE = 0.000112195
K=5: mean = -0.00106825, SD = 0.000498346, SE = 0.00014386
K=10: mean = -0.00249917, SD = 0.00047666, SE = 0.0001376
K=20: mean = -0.00426317, SD = 0.000669796, SE = 0.000193354
K=30: mean = -0.00645875, SD = 0.000655335, SE = 0.000189179
K=50: mean = -0.00931692, SD = 0.000543236, SE = 0.000156819



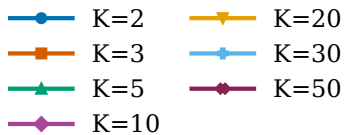
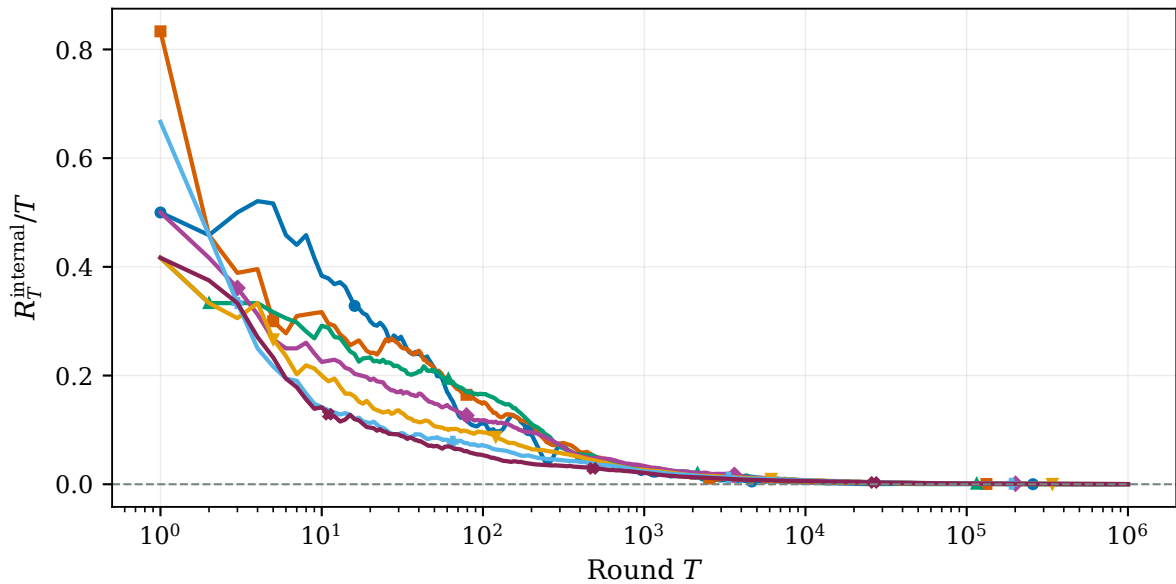
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Full information
Algorithm: BM-OptHedge
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: External regret
View: $R / \sqrt{T}$
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = -0.653833, SD = 0.654497, SE = 0.188937
K=3: mean = -0.763917, SD = 0.388655, SE = 0.112195
K=5: mean = -1.06825, SD = 0.498346, SE = 0.14386
K=10: mean = -2.49917, SD = 0.47666, SE = 0.1376
K=20: mean = -4.26317, SD = 0.669796, SE = 0.193354
K=30: mean = -6.45875, SD = 0.655335, SE = 0.189179
K=50: mean = -9.31692, SD = 0.543236, SE = 0.156819



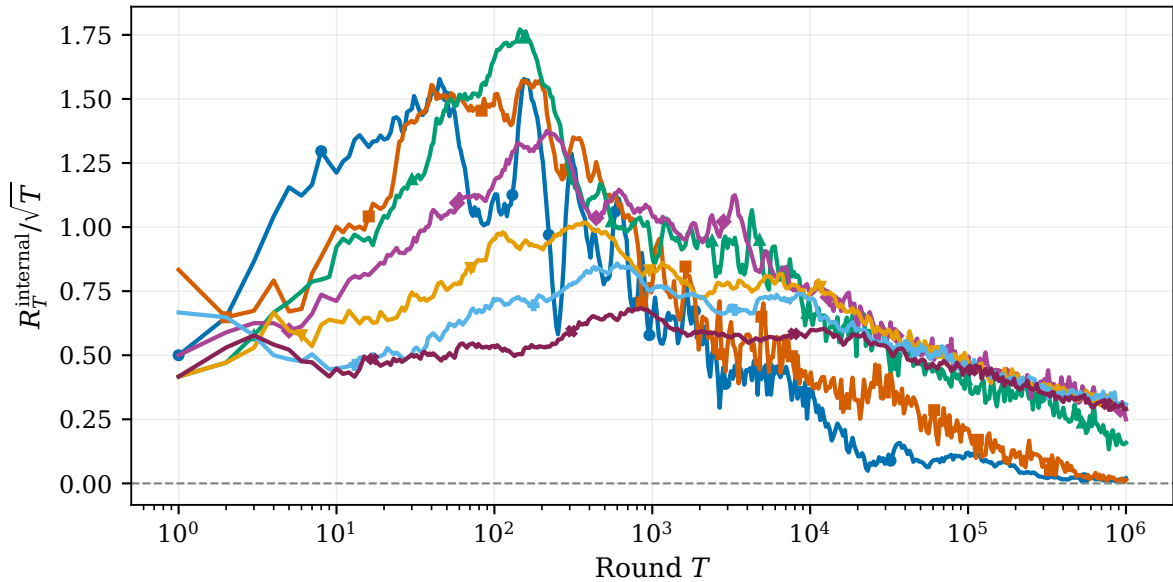
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Full information
Algorithm: BM-OptHedge
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Internal regret
View: R / T
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 2.025e-05, SD = 4.8625e-05, SE = 1.40368e-05
K=3: mean = 1.35e-05, SD = 3.15551e-05, SE = 9.10919e-06
K=5: mean = 0.000158083, SD = 0.000121126, SE = 3.49661e-05
K=10: mean = 0.000249833, SD = 8.82711e-05, SE = 2.54817e-05
K=20: mean = 0.00029175, SD = 7.70916e-05, SE = 2.22544e-05
K=30: mean = 0.00030825, SD = 5.66763e-05, SE = 1.6361e-05
K=50: mean = 0.000290333, SD = 3.05743e-05, SE = 8.82604e-06



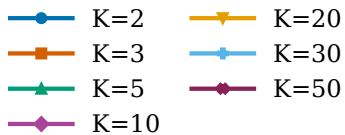
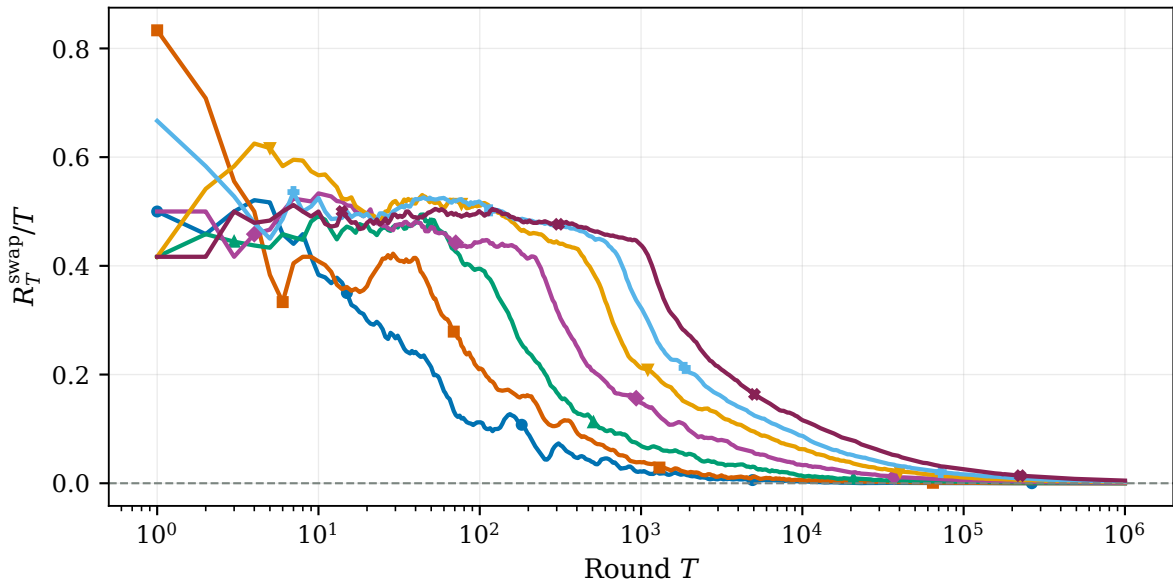
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Full information
Algorithm: BM-OptHedge
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Internal regret
View: $R / \sqrt{T}$
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.02025, SD = 0.048625, SE = 0.0140368
K=3: mean = 0.0135, SD = 0.0315551, SE = 0.00910919
K=5: mean = 0.158083, SD = 0.121126, SE = 0.0349661
K=10: mean = 0.249833, SD = 0.0882711, SE = 0.0254817
K=20: mean = 0.29175, SD = 0.0770916, SE = 0.0222544
K=30: mean = 0.30825, SD = 0.0566763, SE = 0.016361
K=50: mean = 0.290333, SD = 0.0305743, SE = 0.00882604



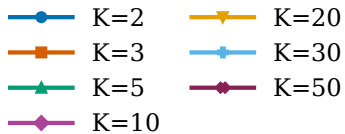
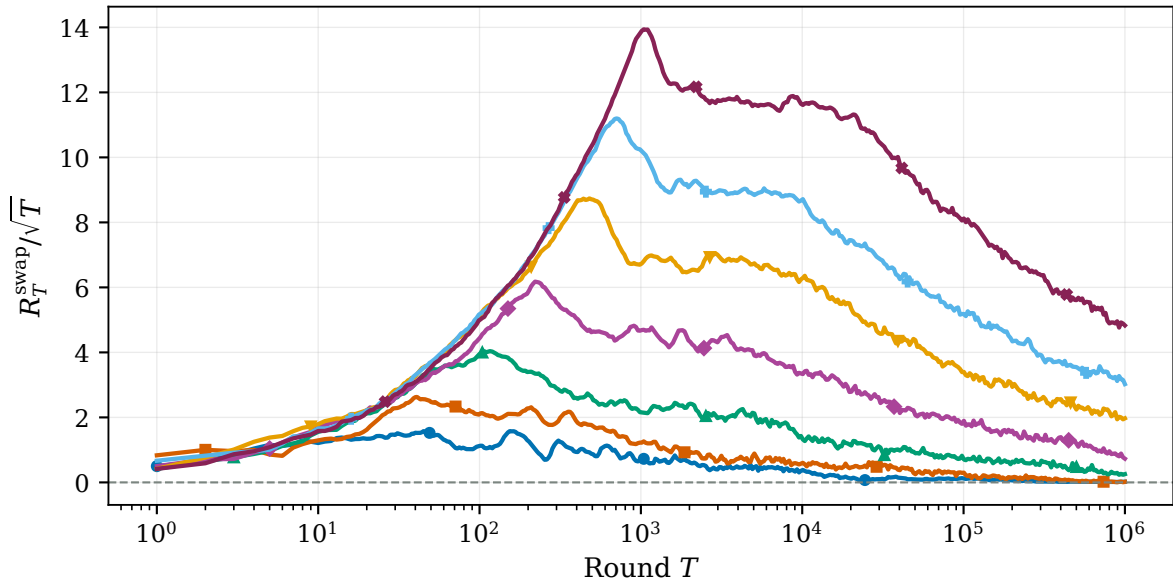
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Full information
Algorithm: BM-OptHedge
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Swap regret
View: R / T
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 2.025e-05, SD = 4.8625e-05, SE = 1.40368e-05
K=3: mean = 1.81667e-05, SD = 4.37469e-05, SE = 1.26286e-05
K=5: mean = 0.000253583, SD = 0.000225814, SE = 6.51868e-05
K=10: mean = 0.000729333, SD = 0.000297851, SE = 8.59821e-05
K=20: mean = 0.00197092, SD = 0.000343091, SE = 9.90419e-05
K=30: mean = 0.00302333, SD = 0.000534752, SE = 0.00015437
K=50: mean = 0.0048295, SD = 0.00051549, SE = 0.000148809



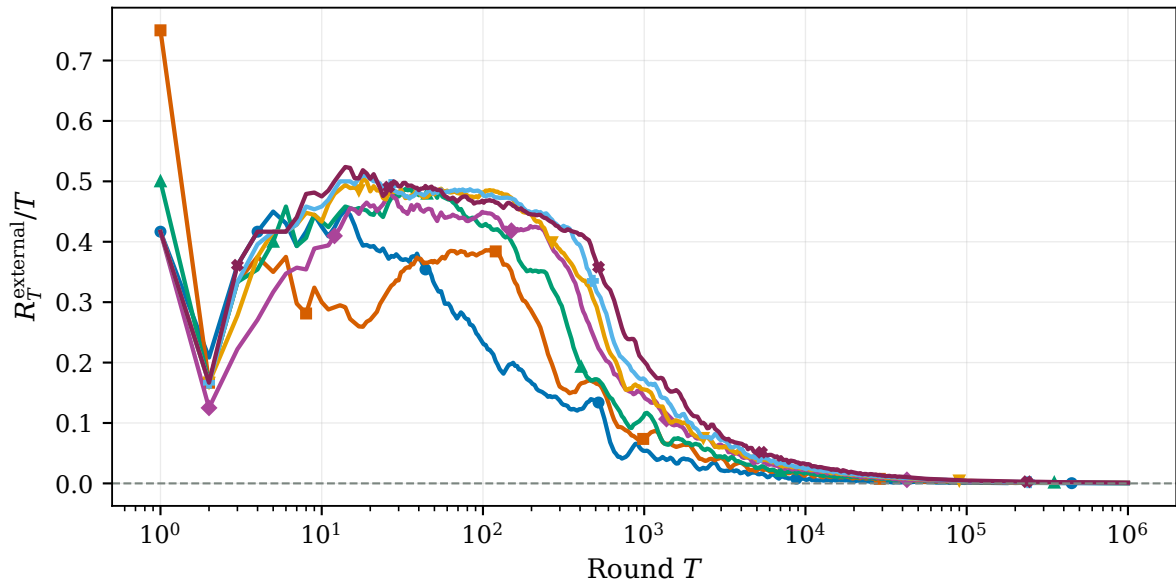
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Full information
Algorithm: BM-OptHedge
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Swap regret
View: $R / \sqrt{T}$
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.02025, SD = 0.048625, SE = 0.0140368
K=3: mean = 0.0181667, SD = 0.0437469, SE = 0.0126286
K=5: mean = 0.253583, SD = 0.225814, SE = 0.0651868
K=10: mean = 0.729333, SD = 0.297851, SE = 0.0859821
K=20: mean = 1.97092, SD = 0.343091, SE = 0.0990419
K=30: mean = 3.02333, SD = 0.534752, SE = 0.15437
K=50: mean = 4.8295, SD = 0.51549, SE = 0.148809



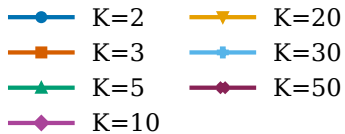
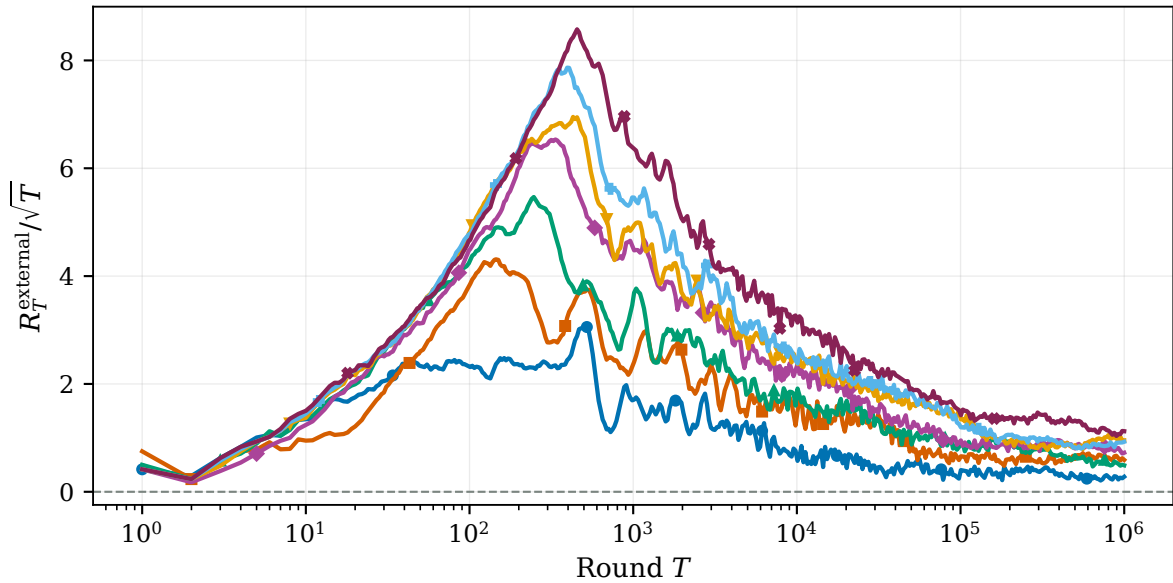
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Full information
Algorithm: Hedge
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: External regret
View: R / T
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.000278833, SD = 0.000409762, SE = 0.000118288
K=3: mean = 0.00059125, SD = 0.000423717, SE = 0.000122317
K=5: mean = 0.00049025, SD = 0.00032575, SE = 9.40359e-05
K=10: mean = 0.0007245, SD = 0.000326643, SE = 9.42938e-05
K=20: mean = 0.0009545, SD = 0.000502, SE = 0.000144915
K=30: mean = 0.000927417, SD = 0.000504499, SE = 0.000145636
K=50: mean = 0.00112367, SD = 0.000390835, SE = 0.000112824



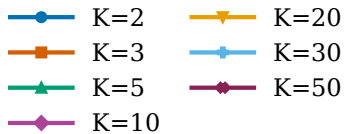
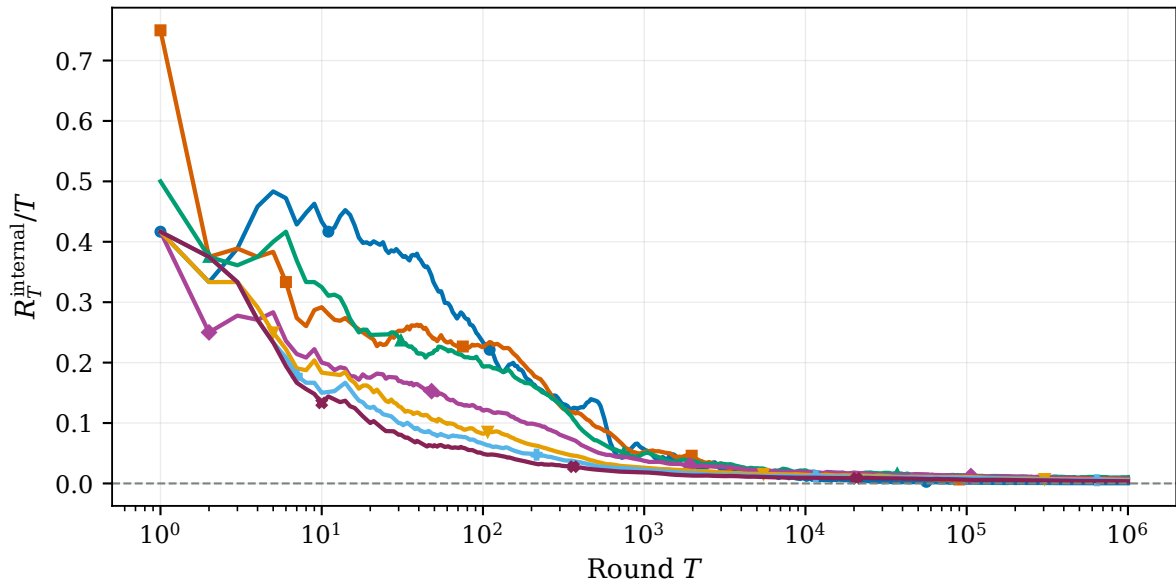
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Full information
Algorithm: Hedge
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: External regret
View: $R / \sqrt{T}$
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.278833, SD = 0.409762, SE = 0.118288
K=3: mean = 0.59125, SD = 0.423717, SE = 0.122317
K=5: mean = 0.49025, SD = 0.32575, SE = 0.0940359
K=10: mean = 0.7245, SD = 0.326643, SE = 0.0942938
K=20: mean = 0.9545, SD = 0.502, SE = 0.144915
K=30: mean = 0.927417, SD = 0.504499, SE = 0.145636
K=50: mean = 1.12367, SD = 0.390835, SE = 0.112824



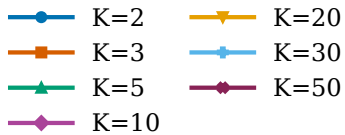
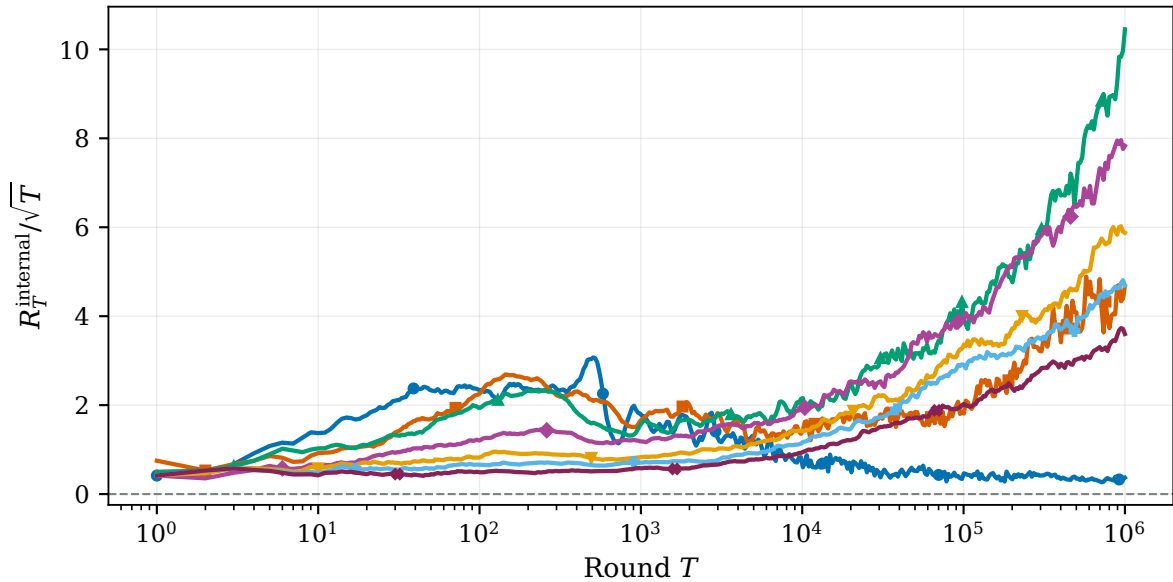
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Full information
Algorithm: Hedge
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Internal regret
View: R / T
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.000366917, SD = 0.000231547, SE = 6.68418e-05
K=3: mean = 0.00467783, SD = 0.00224591, SE = 0.000648339
K=5: mean = 0.0104509, SD = 0.00379174, SE = 0.00109458
K=10: mean = 0.00783042, SD = 0.00149371, SE = 0.000431196
K=20: mean = 0.00587517, SD = 0.000985483, SE = 0.000284484
K=30: mean = 0.004689, SD = 0.00056037, SE = 0.000161765
K=50: mean = 0.00360017, SD = 0.000490659, SE = 0.000141641



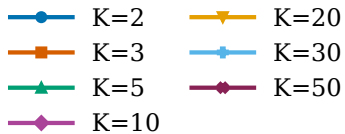
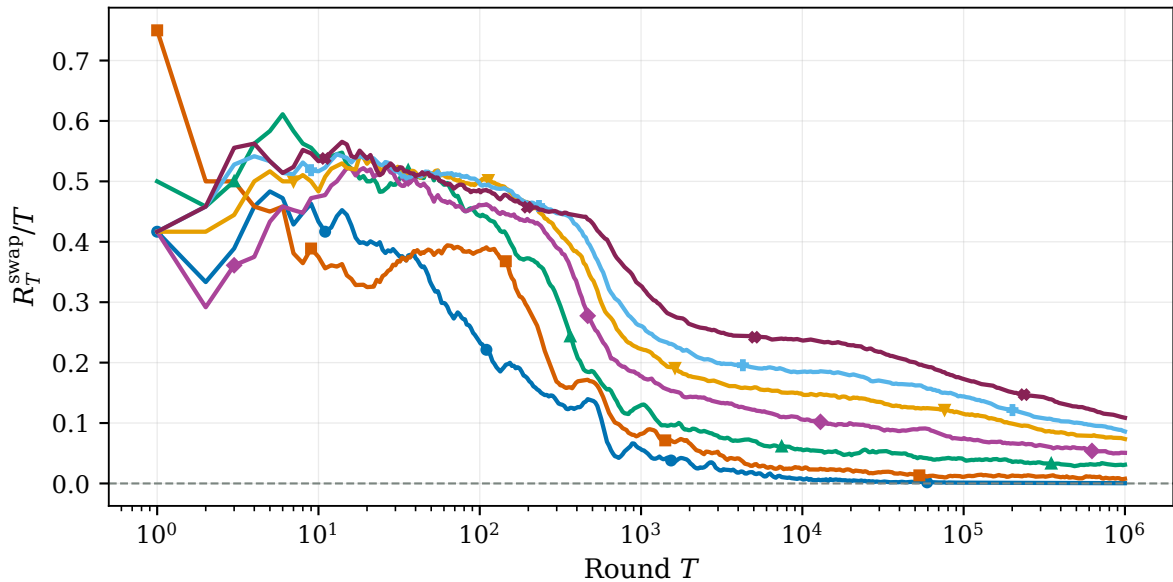
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Full information
Algorithm: Hedge
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Internal regret
View: $R / \sqrt{T}$
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.366917, SD = 0.231547, SE = 0.0668418
K=3: mean = 4.67783, SD = 2.24591, SE = 0.648339
K=5: mean = 10.4509, SD = 3.79174, SE = 1.09458
K=10: mean = 7.83042, SD = 1.49371, SE = 0.431196
K=20: mean = 5.87517, SD = 0.985483, SE = 0.284484
K=30: mean = 4.689, SD = 0.56037, SE = 0.161765
K=50: mean = 3.60017, SD = 0.490659, SE = 0.141641



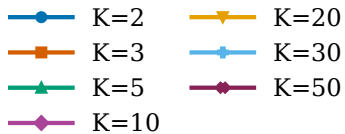
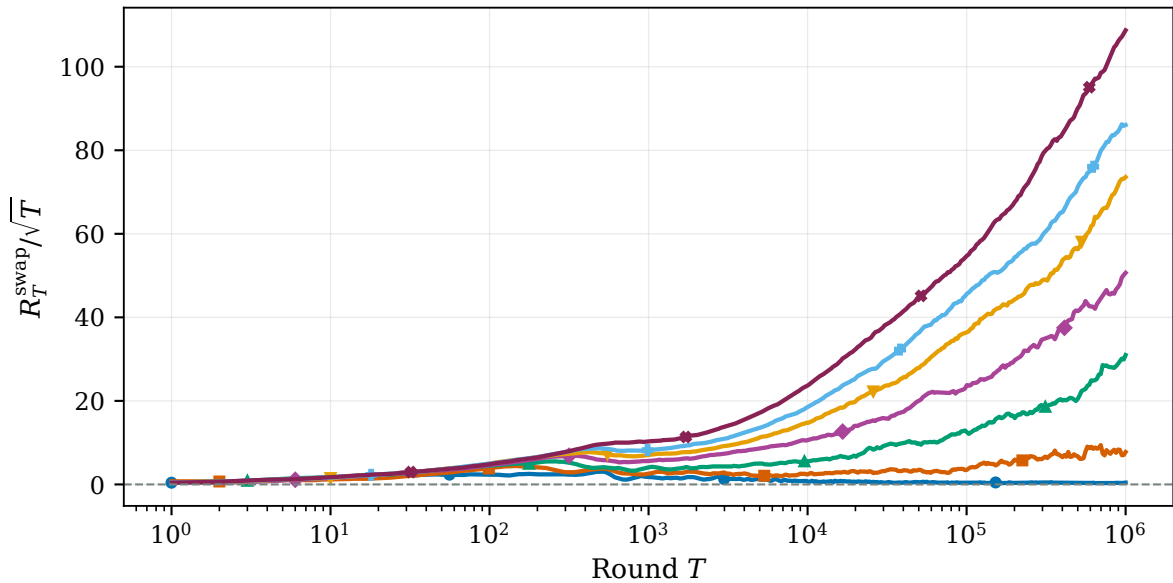
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Full information
Algorithm: Hedge
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Swap regret
View: R / T
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.000434667, SD = 0.000348112, SE = 0.000100491
K=3: mean = 0.00775867, SD = 0.00446792, SE = 0.00128978
K=5: mean = 0.0309706, SD = 0.00995847, SE = 0.00287476
K=10: mean = 0.0506502, SD = 0.00808444, SE = 0.00233378
K=20: mean = 0.0735651, SD = 0.00576812, SE = 0.00166511
K=30: mean = 0.0860448, SD = 0.00482407, SE = 0.00139259
K=50: mean = 0.108715, SD = 0.00433366, SE = 0.00125102



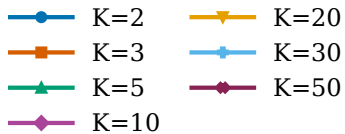
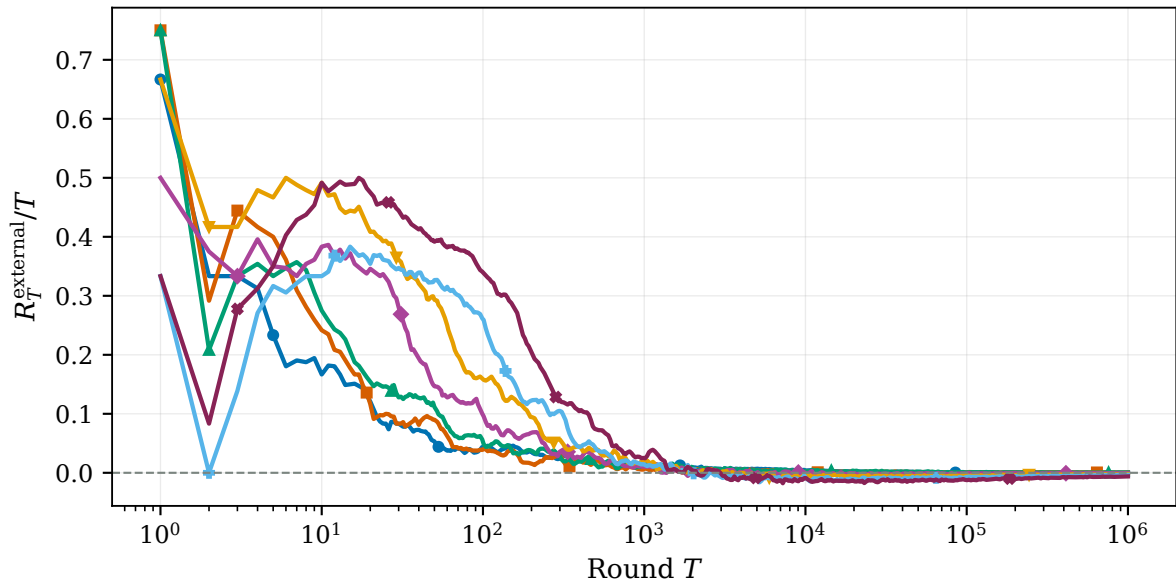
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Full information
Algorithm: Hedge
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Swap regret
View: $R / \sqrt{T}$
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.434667, SD = 0.348112, SE = 0.100491
K=3: mean = 7.75867, SD = 4.46792, SE = 1.28978
K=5: mean = 30.9706, SD = 9.95847, SE = 2.87476
K=10: mean = 50.6502, SD = 8.08444, SE = 2.33378
K=20: mean = 73.5651, SD = 5.76812, SE = 1.66511
K=30: mean = 86.0448, SD = 4.82407, SE = 1.39259
K=50: mean = 108.715, SD = 4.33366, SE = 1.25102



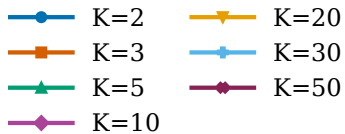
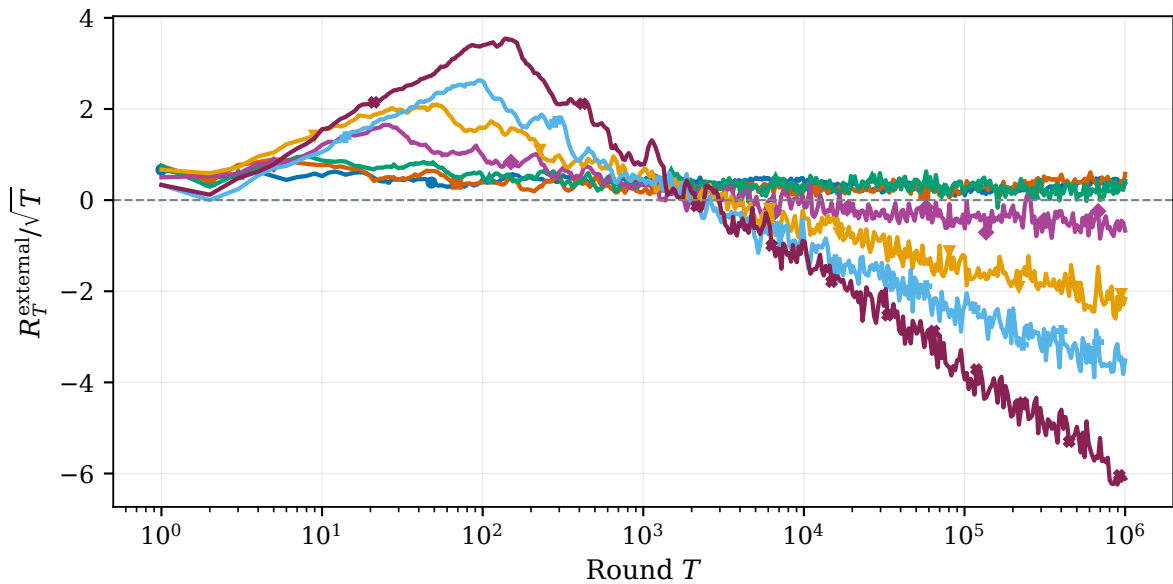
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Full information
Algorithm: Ito-Hedge
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: External regret
View: R / T
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.000367667, SD = 0.000279906, SE = 8.08019e-05
K=3: mean = 0.000576333, SD = 0.000576892, SE = 0.000166534
K=5: mean = 0.000421417, SD = 0.000445002, SE = 0.000128461
K=10: mean = -0.000669917, SD = 0.00050081, SE = 0.000144571
K=20: mean = -0.0021645, SD = 0.00077929, SE = 0.000224962
K=30: mean = -0.0035185, SD = 0.000658683, SE = 0.000190145
K=50: mean = -0.00611258, SD = 0.00118647, SE = 0.000342506



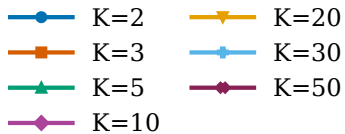
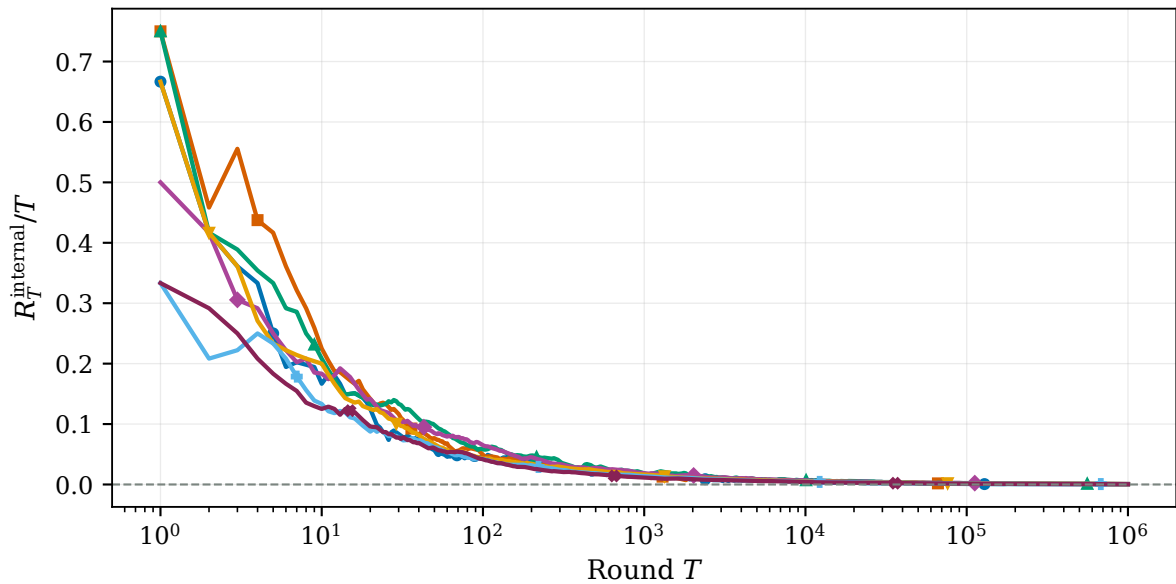
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Full information
Algorithm: Ito-Hedge
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: External regret
View: $R / \sqrt{T}$
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.367667, SD = 0.279906, SE = 0.0808019
K=3: mean = 0.576333, SD = 0.576892, SE = 0.166534
K=5: mean = 0.421417, SD = 0.445002, SE = 0.128461
K=10: mean = -0.669917, SD = 0.50081, SE = 0.144571
K=20: mean = -2.1645, SD = 0.77929, SE = 0.224962
K=30: mean = -3.5185, SD = 0.658683, SE = 0.190145
K=50: mean = -6.11258, SD = 1.18647, SE = 0.342506



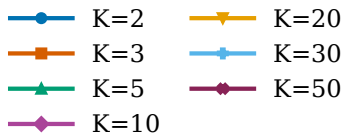
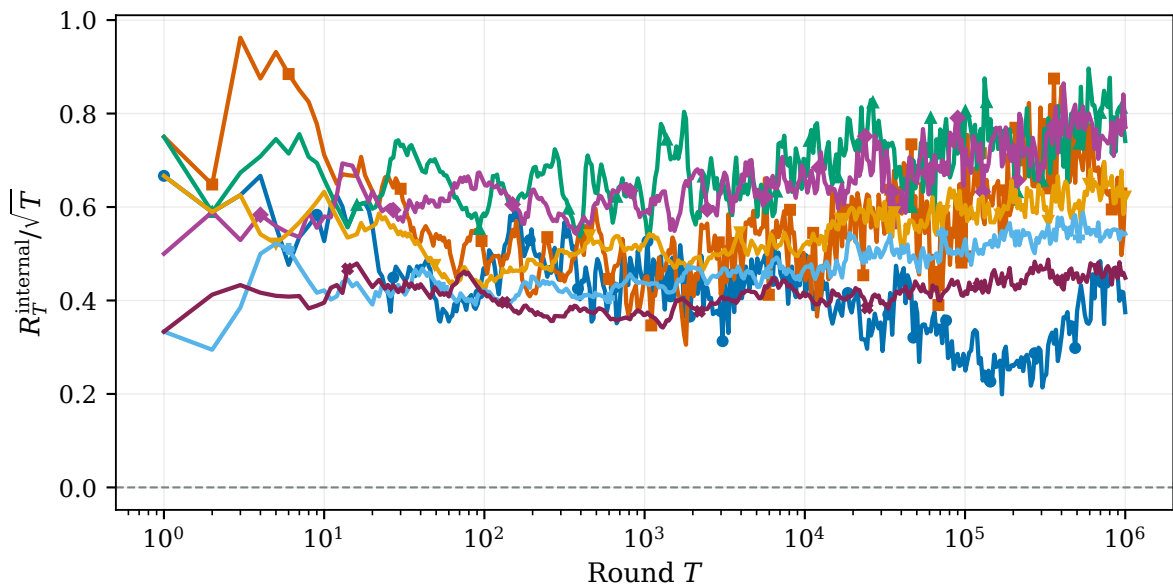
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Full information
Algorithm: Ito-Hedge
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Internal regret
View: R / T
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.000375, SD = 0.000267772, SE = 7.72992e-05
K=3: mean = 0.000637333, SD = 0.000337767, SE = 9.75048e-05
K=5: mean = 0.00074125, SD = 0.000219716, SE = 6.34264e-05
K=10: mean = 0.000771583, SD = 0.000209516, SE = 6.0482e-05
K=20: mean = 0.0006225, SD = 9.87085e-05, SE = 2.84947e-05
K=30: mean = 0.000542667, SD = 7.02571e-05, SE = 2.02815e-05
K=50: mean = 0.000448667, SD = 7.1731e-05, SE = 2.07069e-05



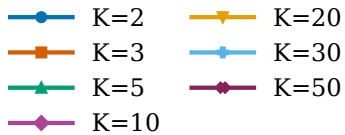
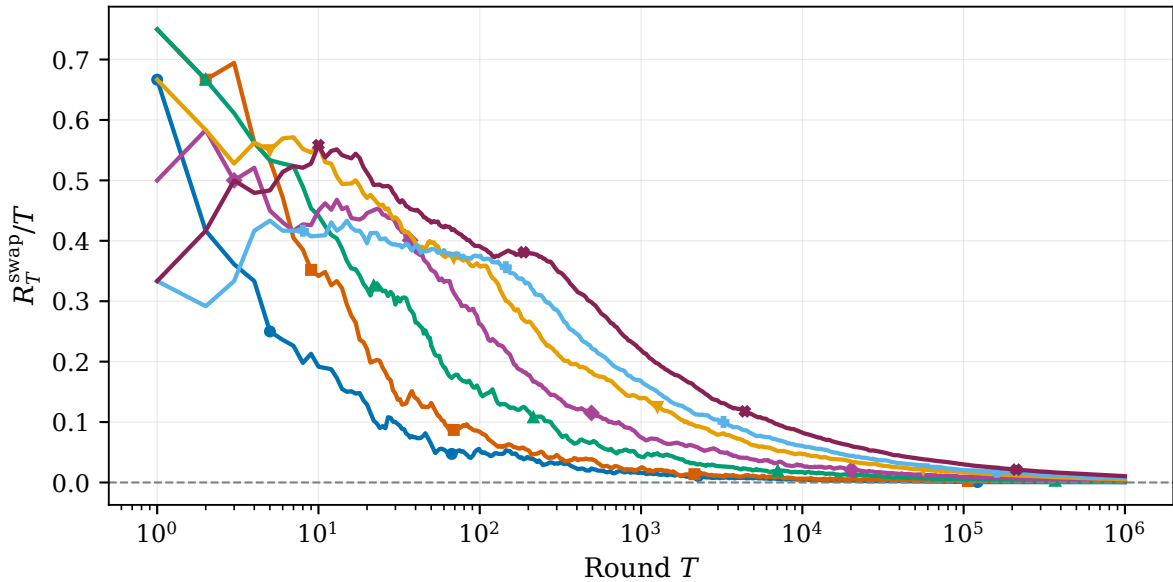
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Full information
Algorithm: Ito-Hedge
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Internal regret
View: $R / \sqrt{T}$
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.375, SD = 0.267772, SE = 0.0772992
K=3: mean = 0.637333, SD = 0.337767, SE = 0.0975048
K=5: mean = 0.74125, SD = 0.219716, SE = 0.0634264
K=10: mean = 0.771583, SD = 0.209516, SE = 0.060482
K=20: mean = 0.6225, SD = 0.0987085, SE = 0.0284947
K=30: mean = 0.542667, SD = 0.0702571, SE = 0.0202815
K=50: mean = 0.448667, SD = 0.071731, SE = 0.0207069



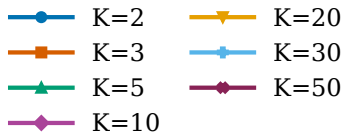
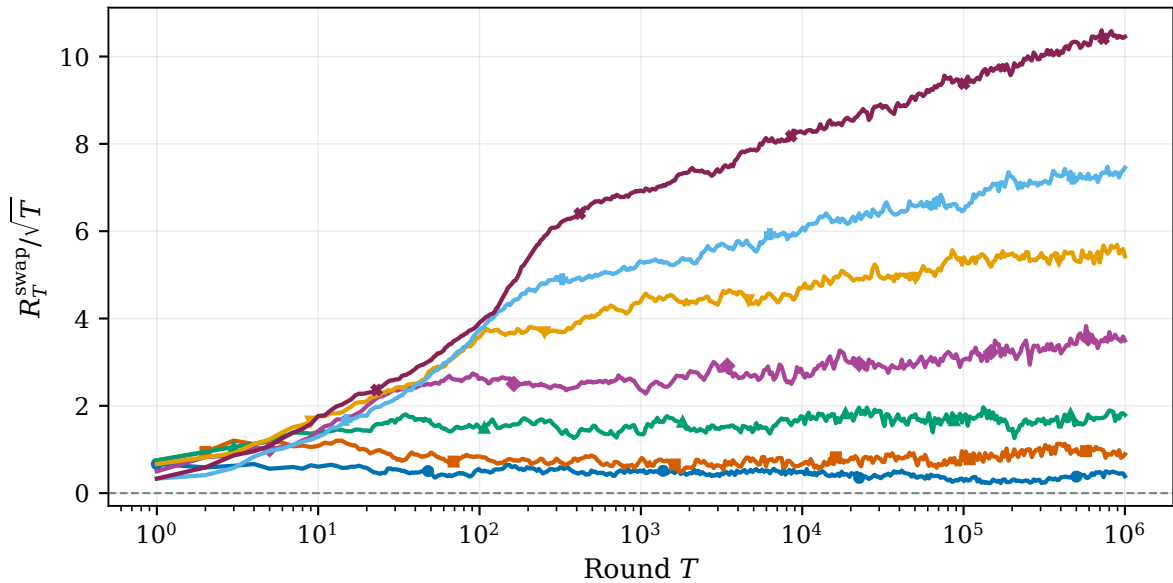
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Full information
Algorithm: Ito-Hedge
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Swap regret
View: R / T
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.000386, SD = 0.000259392, SE = 7.48801e-05
K=3: mean = 0.000893167, SD = 0.000562902, SE = 0.000162496
K=5: mean = 0.00179033, SD = 0.000433119, SE = 0.000125031
K=10: mean = 0.00349483, SD = 0.000596753, SE = 0.000172268
K=20: mean = 0.00542342, SD = 0.000619034, SE = 0.0001787
K=30: mean = 0.00745183, SD = 0.000478928, SE = 0.000138255
K=50: mean = 0.0104534, SD = 0.000511925, SE = 0.00014778



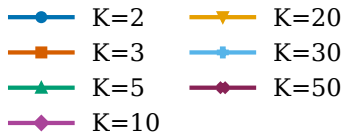
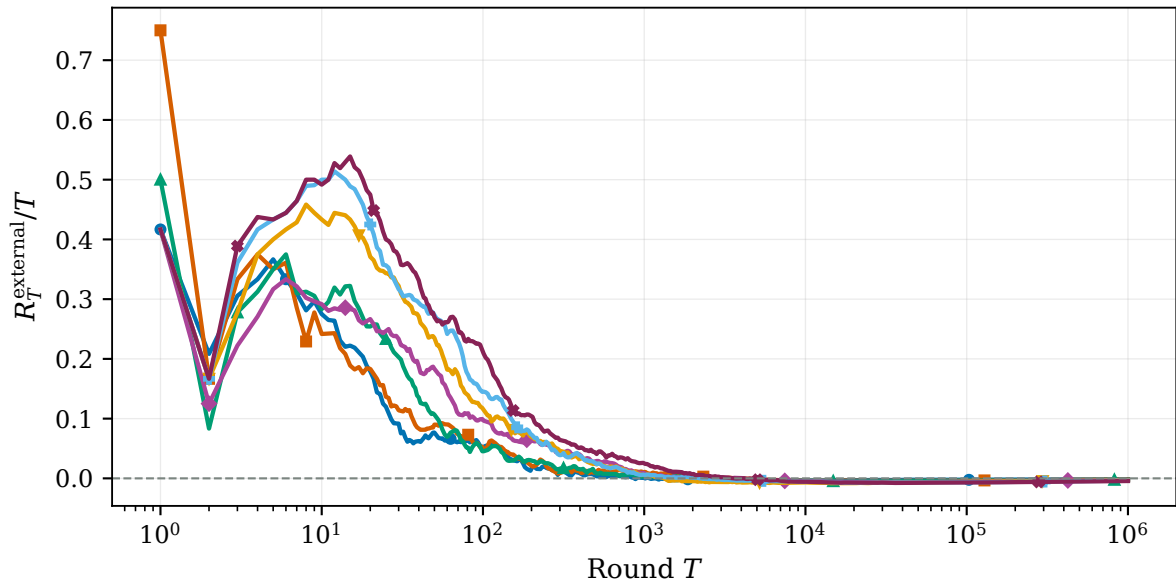
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Full information
Algorithm: Ito-Hedge
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Swap regret
View: $R / \sqrt{T}$
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.386, SD = 0.259392, SE = 0.0748801
K=3: mean = 0.893167, SD = 0.562902, SE = 0.162496
K=5: mean = 1.79033, SD = 0.433119, SE = 0.125031
K=10: mean = 3.49483, SD = 0.596753, SE = 0.172268
K=20: mean = 5.42342, SD = 0.619034, SE = 0.1787
K=30: mean = 7.45183, SD = 0.478928, SE = 0.138255
K=50: mean = 10.4534, SD = 0.511925, SE = 0.14778



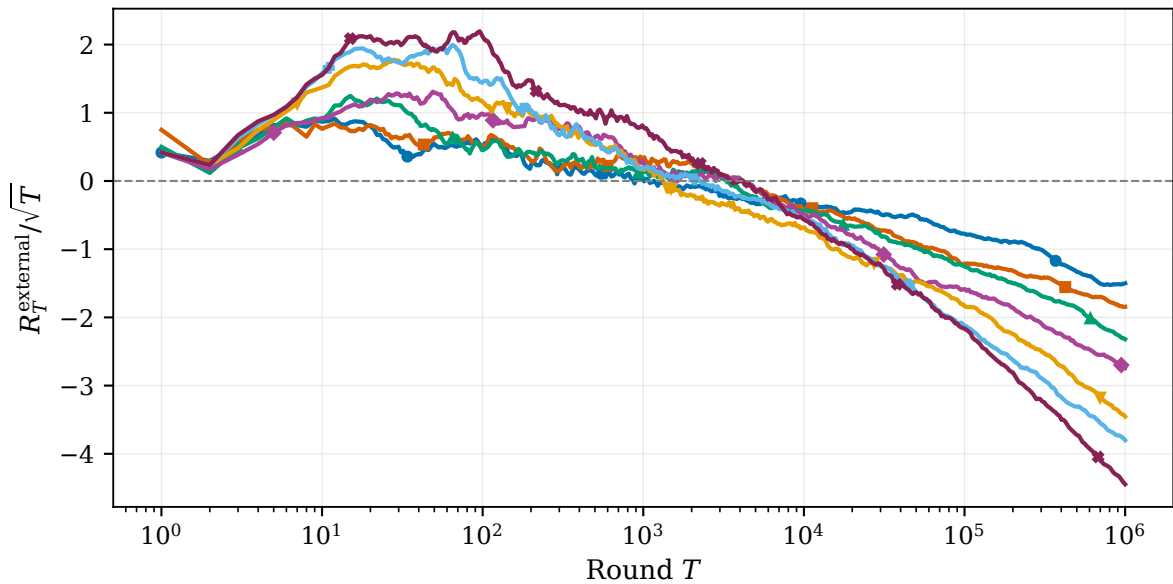
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Full information
Algorithm: OptHedge
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: External regret
View: R / T
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = -0.00149958, SD = 0.000647914, SE = 0.000187037
K=3: mean = -0.00184475, SD = 0.000432252, SE = 0.00012478
K=5: mean = -0.00231975, SD = 0.000547053, SE = 0.00015792
K=10: mean = -0.00274242, SD = 0.00049422, SE = 0.000142669
K=20: mean = -0.00345508, SD = 0.000486174, SE = 0.000140346
K=30: mean = -0.0038015, SD = 0.000395642, SE = 0.000114212
K=50: mean = -0.0044465, SD = 0.000483935, SE = 0.0001397



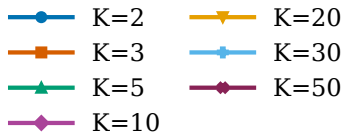
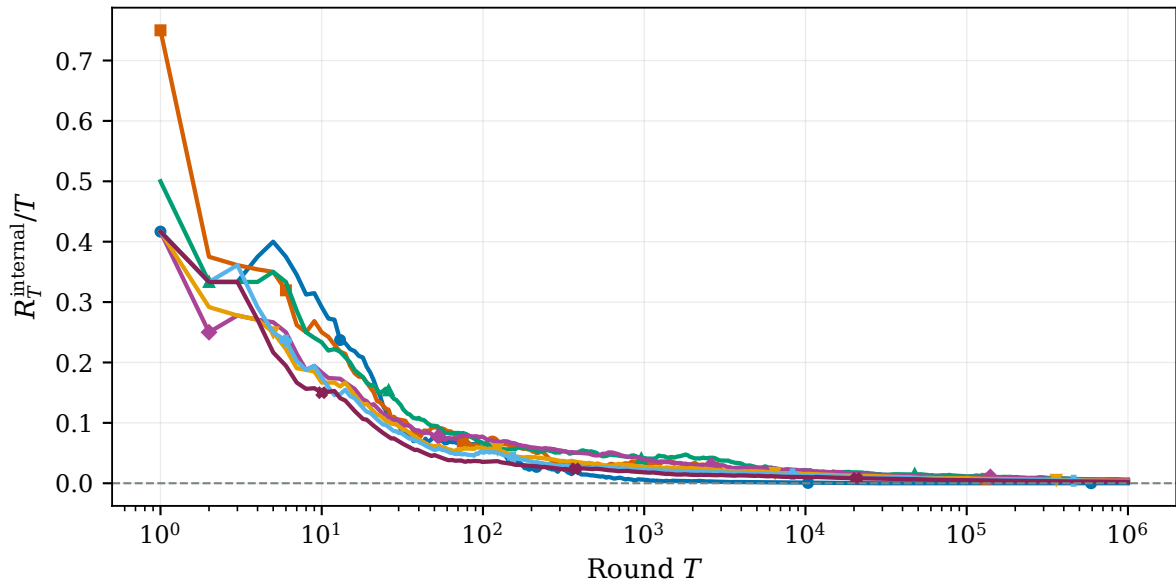
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Full information
Algorithm: OptHedge
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: External regret
View: $R / \sqrt{T}$
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = -1.49958, SD = 0.647914, SE = 0.187037
K=3: mean = -1.84475, SD = 0.432252, SE = 0.12478
K=5: mean = -2.31975, SD = 0.547053, SE = 0.15792
K=10: mean = -2.74242, SD = 0.49422, SE = 0.142669
K=20: mean = -3.45508, SD = 0.486174, SE = 0.140346
K=30: mean = -3.8015, SD = 0.395642, SE = 0.114212
K=50: mean = -4.4465, SD = 0.483935, SE = 0.1397



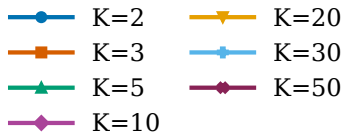
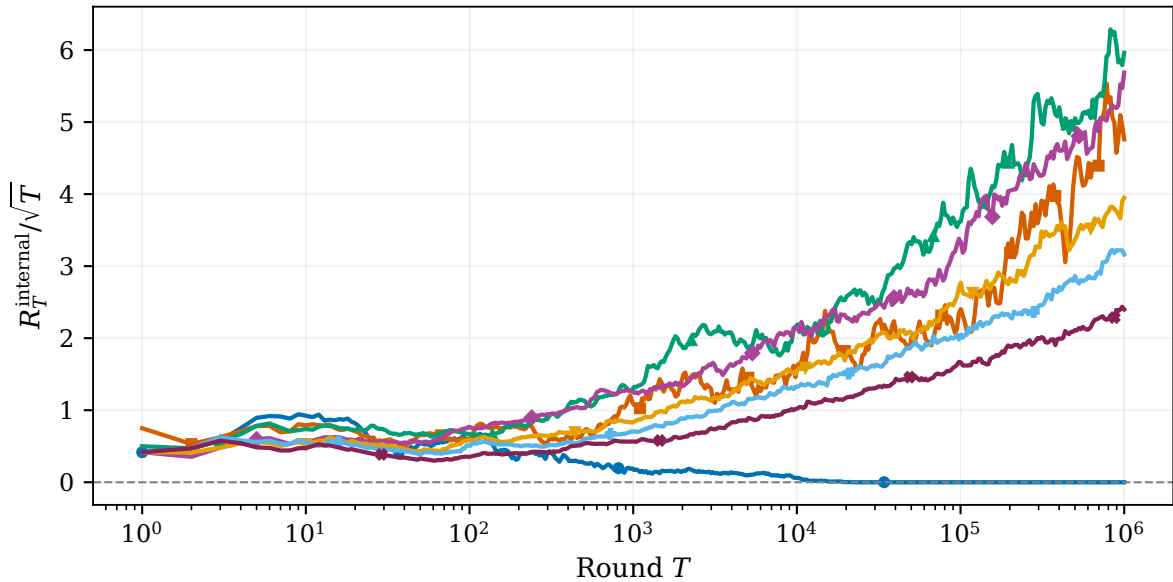
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Full information
Algorithm: OptHedge
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Internal regret
View: R / T
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0, SD = 0, SE = 0
K=3: mean = 0.00476, SD = 0.0029225, SE = 0.000843654
K=5: mean = 0.0059615, SD = 0.00197131, SE = 0.000569069
K=10: mean = 0.00568767, SD = 0.000769671, SE = 0.000222185
K=20: mean = 0.00394867, SD = 0.000536064, SE = 0.000154748
K=30: mean = 0.0031625, SD = 0.000294047, SE = 8.48841e-05
K=50: mean = 0.00239858, SD = 0.000218376, SE = 6.30398e-05



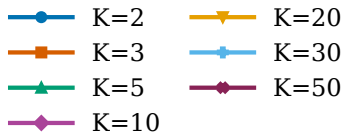
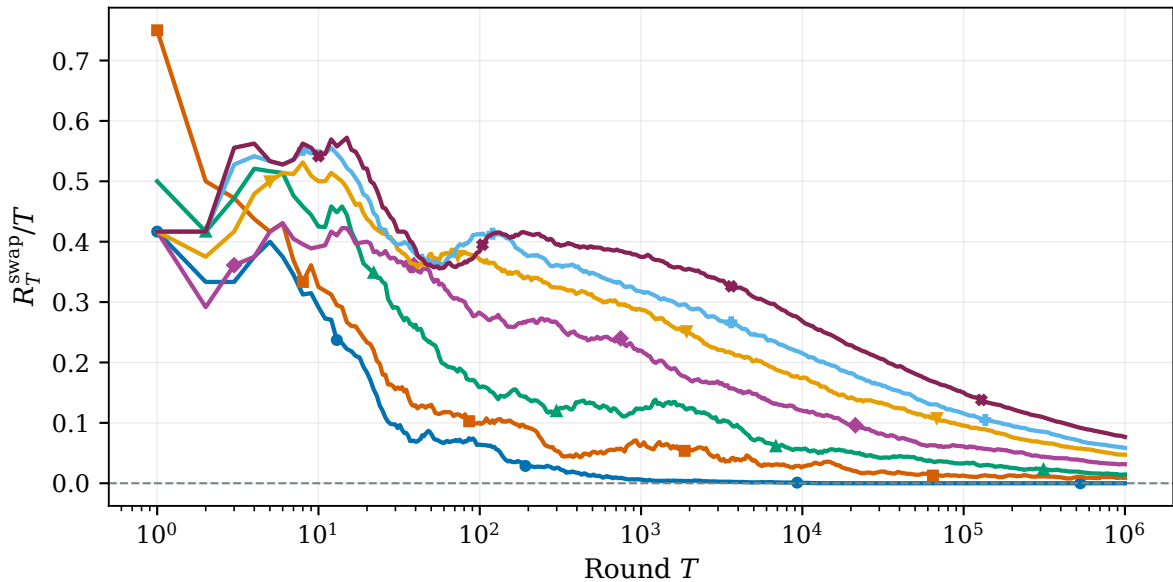
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Full information
Algorithm: OptHedge
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Internal regret
View: $R / \sqrt{T}$
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0, SD = 0, SE = 0
K=3: mean = 4.76, SD = 2.9225, SE = 0.843654
K=5: mean = 5.9615, SD = 1.97131, SE = 0.569069
K=10: mean = 5.68767, SD = 0.769671, SE = 0.222185
K=20: mean = 3.94867, SD = 0.536064, SE = 0.154748
K=30: mean = 3.1625, SD = 0.294047, SE = 0.0848841
K=50: mean = 2.39858, SD = 0.218376, SE = 0.0630398



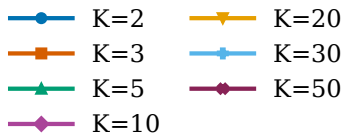
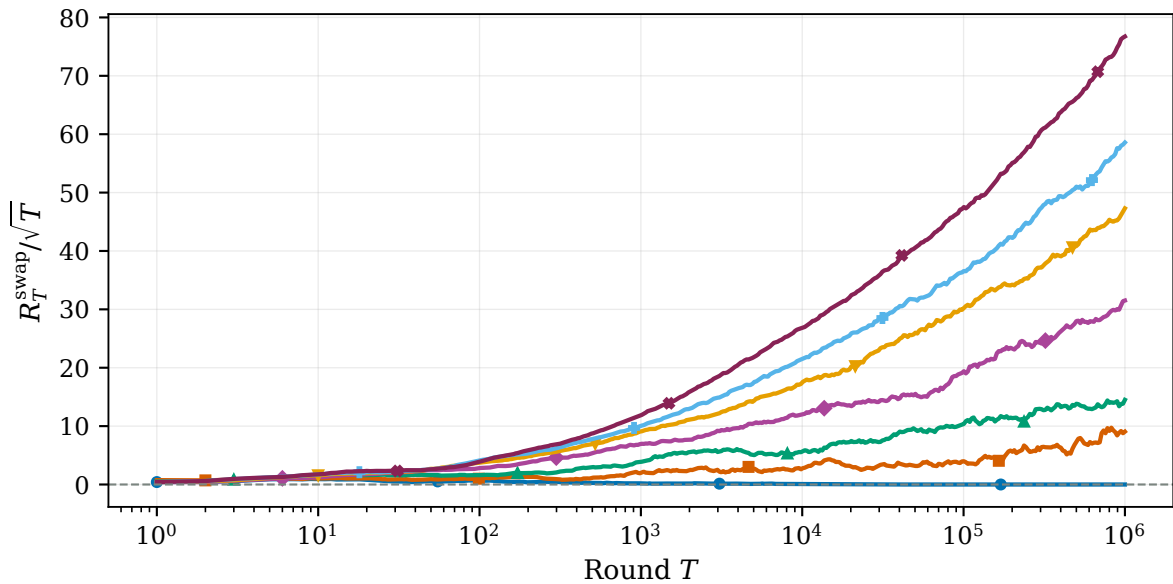
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Full information
Algorithm: OptHedge
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Swap regret
View: R / T
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0, SD = 0, SE = 0
K=3: mean = 0.00905317, SD = 0.00789552, SE = 0.00227924
K=5: mean = 0.0144973, SD = 0.00410148, SE = 0.00118399
K=10: mean = 0.0315117, SD = 0.00529378, SE = 0.00152818
K=20: mean = 0.0472968, SD = 0.00339307, SE = 0.000979494
K=30: mean = 0.0585762, SD = 0.00378225, SE = 0.00109184
K=50: mean = 0.0767349, SD = 0.0039632, SE = 0.00114408



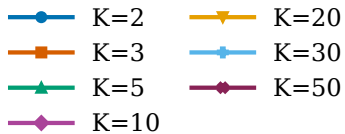
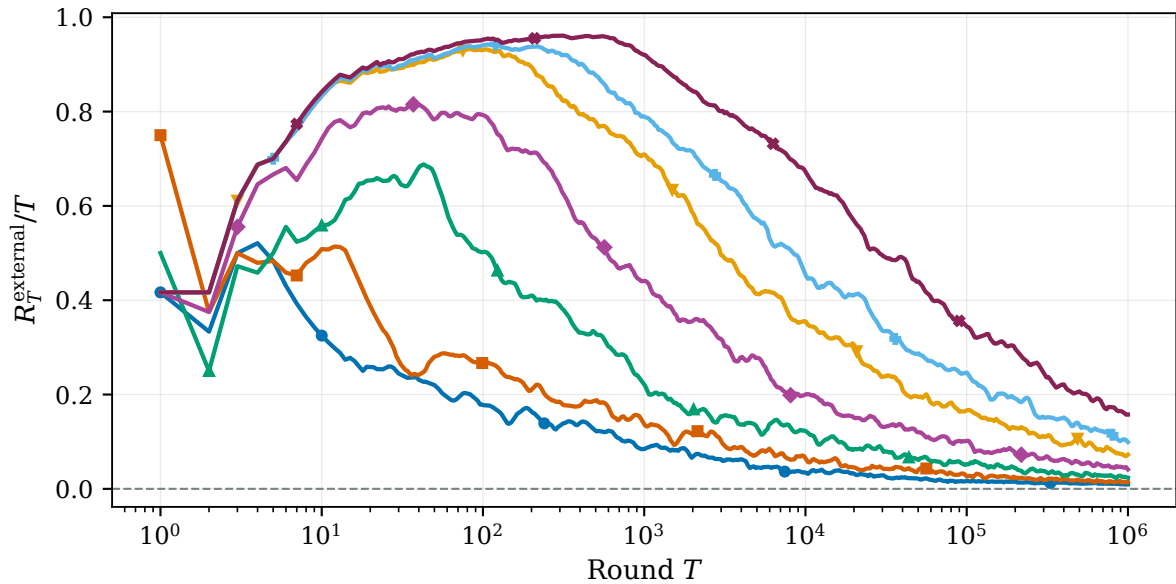
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Full information
Algorithm: OptHedge
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Swap regret
View: $R / \sqrt{T}$
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0, SD = 0, SE = 0
K=3: mean = 9.05317, SD = 7.89552, SE = 2.27924
K=5: mean = 14.4973, SD = 4.10148, SE = 1.18399
K=10: mean = 31.5117, SD = 5.29378, SE = 1.52818
K=20: mean = 47.2968, SD = 3.39307, SE = 0.979494
K=30: mean = 58.5762, SD = 3.78225, SE = 1.09184
K=50: mean = 76.7349, SD = 3.9632, SE = 1.14408



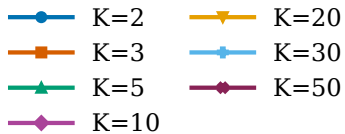
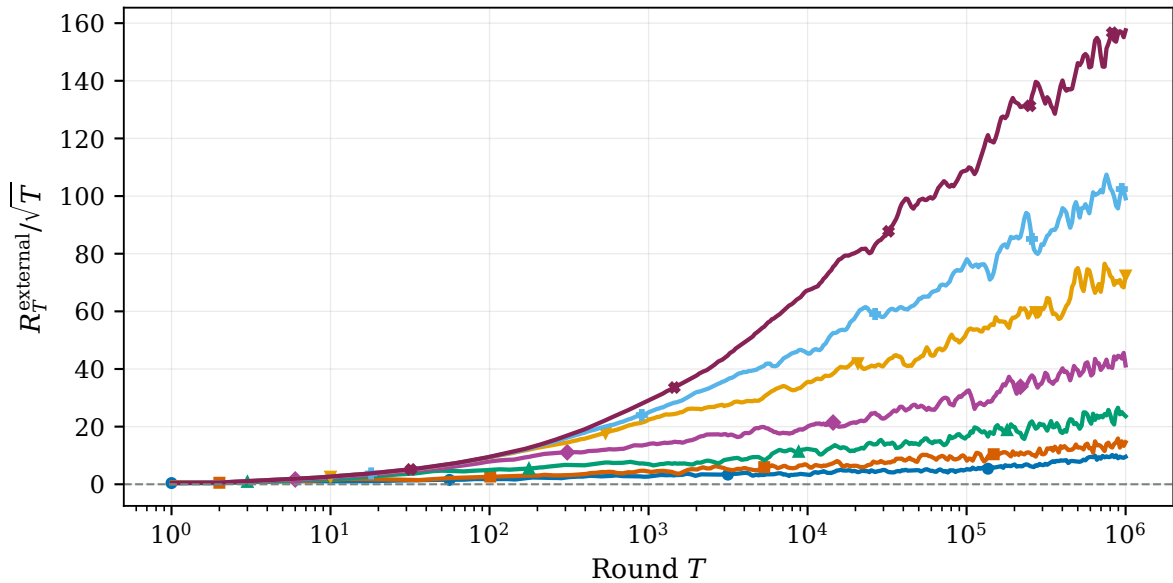
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Full information
Algorithm: RM
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: External regret
View: R / T
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.00954458, SD = 0.00341066, SE = 0.000984573
K=3: mean = 0.0146044, SD = 0.00480609, SE = 0.0013874
K=5: mean = 0.0236186, SD = 0.00676298, SE = 0.0019523
K=10: mean = 0.0412372, SD = 0.0104722, SE = 0.00302308
K=20: mean = 0.0724448, SD = 0.0120501, SE = 0.00347855
K=30: mean = 0.0992312, SD = 0.0136522, SE = 0.00394105
K=50: mean = 0.157515, SD = 0.0155822, SE = 0.00449819



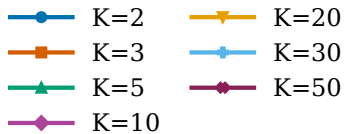
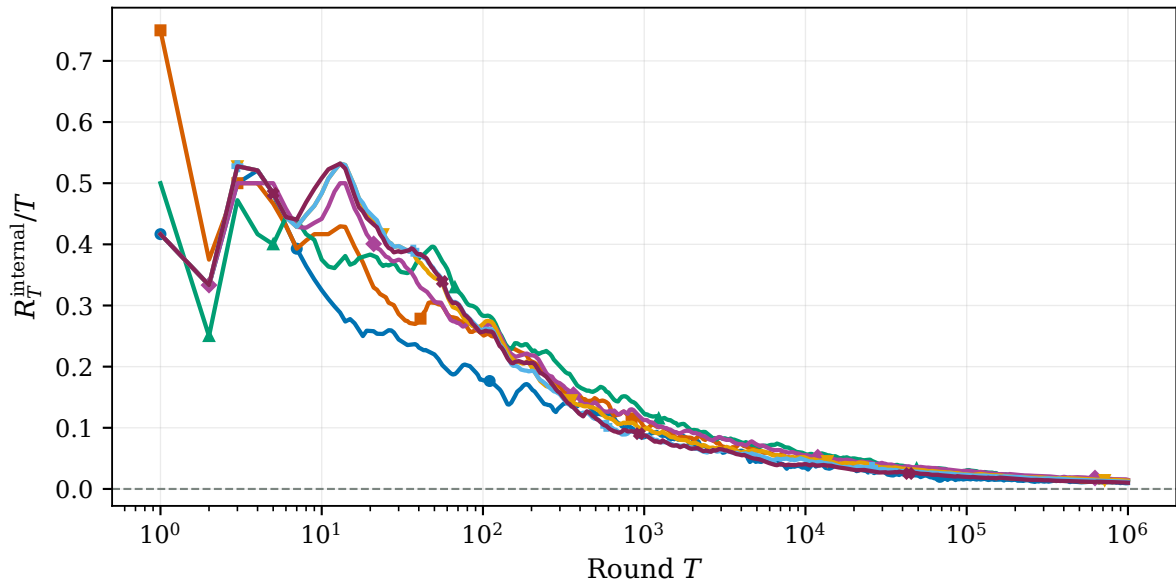
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Full information
Algorithm: RM
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: External regret
View: $R / \sqrt{T}$
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 9.54458, SD = 3.41066, SE = 0.984573
K=3: mean = 14.6044, SD = 4.80609, SE = 1.3874
K=5: mean = 23.6186, SD = 6.76298, SE = 1.9523
K=10: mean = 41.2372, SD = 10.4722, SE = 3.02308
K=20: mean = 72.4447, SD = 12.0501, SE = 3.47855
K=30: mean = 99.2313, SD = 13.6522, SE = 3.94105
K=50: mean = 157.515, SD = 15.5822, SE = 4.49819



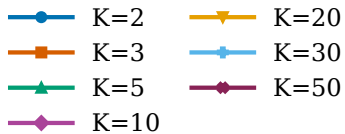
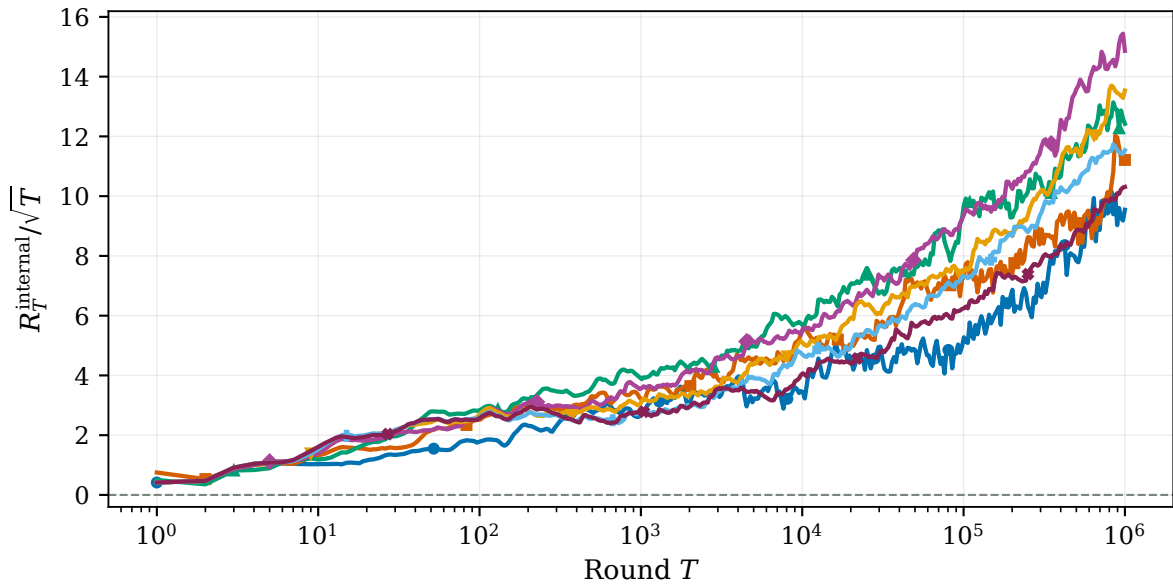
Experiment Information

Figure:	Action-space comparison
Environment:	Historical-frequency
Actions:	2, 3, 5, 10, 20, 30, 50
Feedback:	Full information
Algorithm:	RM
Horizon:	1,000,000
Replicates:	12
Seed:	42
Metric:	Internal regret
View:	R / T
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2:	mean = 0.00954458, SD = 0.00341066, SE = 0.000984573
K=3:	mean = 0.0112128, SD = 0.00275019, SE = 0.00079391
K=5:	mean = 0.0124222, SD = 0.00198889, SE = 0.000574144
K=10:	mean = 0.014864, SD = 0.00232601, SE = 0.000671462
K=20:	mean = 0.0135349, SD = 0.00183288, SE = 0.000529106
K=30:	mean = 0.0115362, SD = 0.000643155, SE = 0.000185663
K=50:	mean = 0.0103103, SD = 0.000683008, SE = 0.000197168



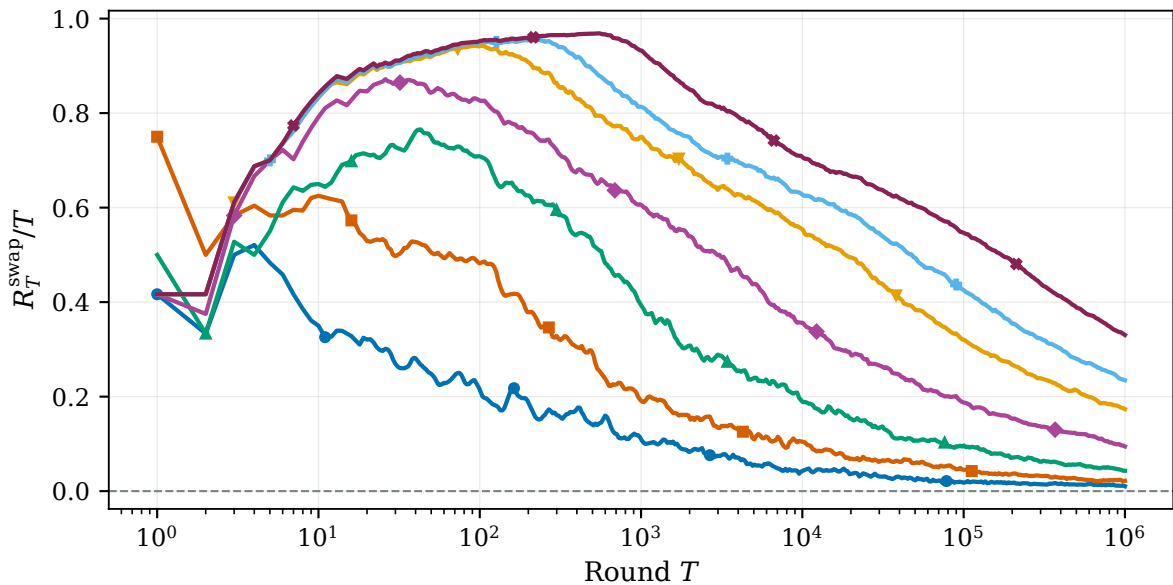
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Full information
Algorithm: RM
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Internal regret
View: $R / \sqrt{T}$
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 9.54458, SD = 3.41066, SE = 0.984573
K=3: mean = 11.2127, SD = 2.75019, SE = 0.79391
K=5: mean = 12.4222, SD = 1.98889, SE = 0.574144
K=10: mean = 14.864, SD = 2.32601, SE = 0.671462
K=20: mean = 13.5349, SD = 1.83288, SE = 0.529106
K=30: mean = 11.5363, SD = 0.643155, SE = 0.185663
K=50: mean = 10.3103, SD = 0.683008, SE = 0.197168



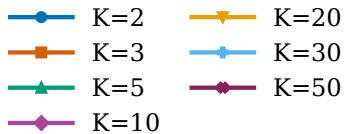
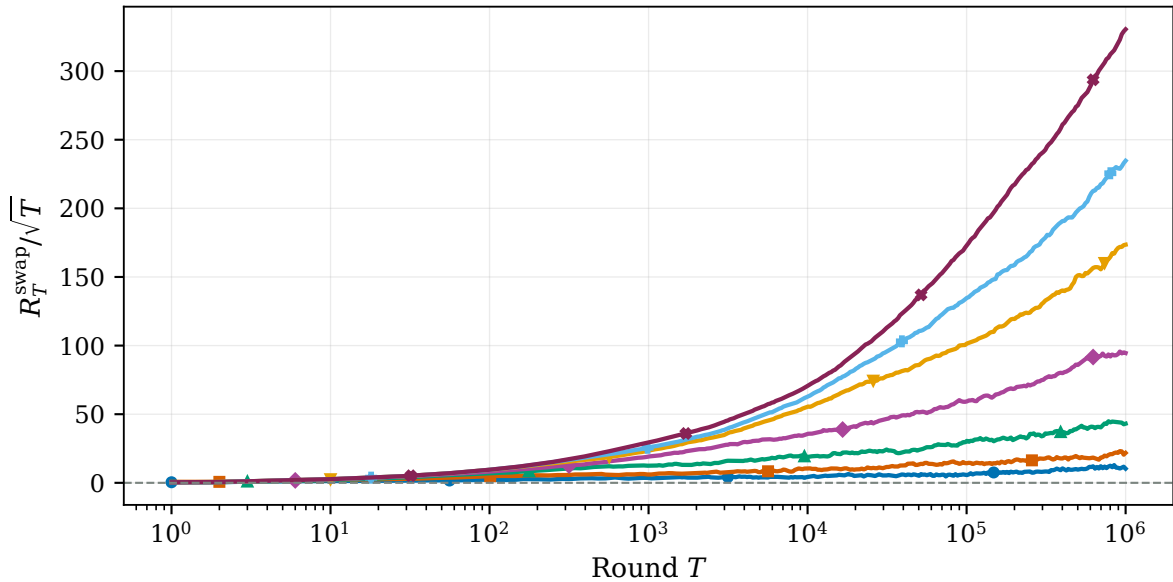
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Full information
Algorithm: RM
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Swap regret
View: R / T
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.0103254, SD = 0.00321766, SE = 0.000928858
K=3: mean = 0.0217964, SD = 0.00499177, SE = 0.001441
K=5: mean = 0.0431142, SD = 0.00558773, SE = 0.00161304
K=10: mean = 0.0945782, SD = 0.00788139, SE = 0.00227516
K=20: mean = 0.173528, SD = 0.00922116, SE = 0.00266192
K=30: mean = 0.234661, SD = 0.0101978, SE = 0.00294384
K=50: mean = 0.330452, SD = 0.00490923, SE = 0.00141717



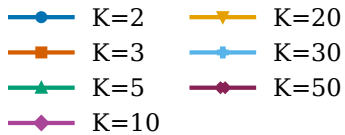
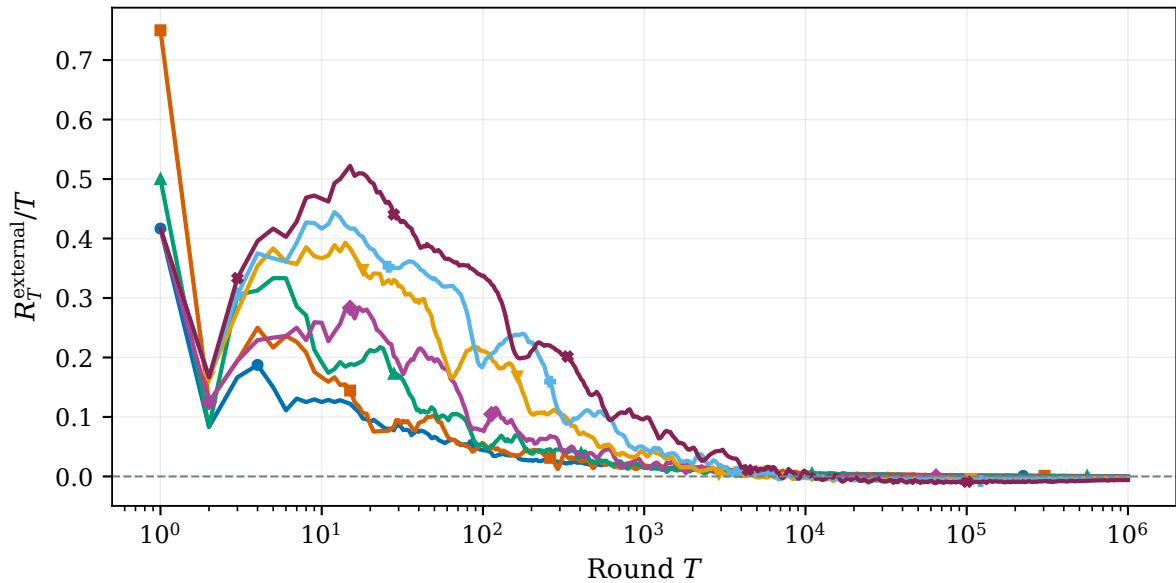
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Full information
Algorithm: RM
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Swap regret
View: $R / \sqrt{T}$
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 10.3254, SD = 3.21766, SE = 0.928858
K=3: mean = 21.7964, SD = 4.99177, SE = 1.441
K=5: mean = 43.1142, SD = 5.58773, SE = 1.61304
K=10: mean = 94.5783, SD = 7.88139, SE = 2.27516
K=20: mean = 173.528, SD = 9.22116, SE = 2.66192
K=30: mean = 234.661, SD = 10.1978, SE = 2.94384
K=50: mean = 330.452, SD = 4.90923, SE = 1.41717



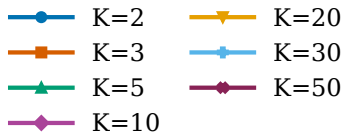
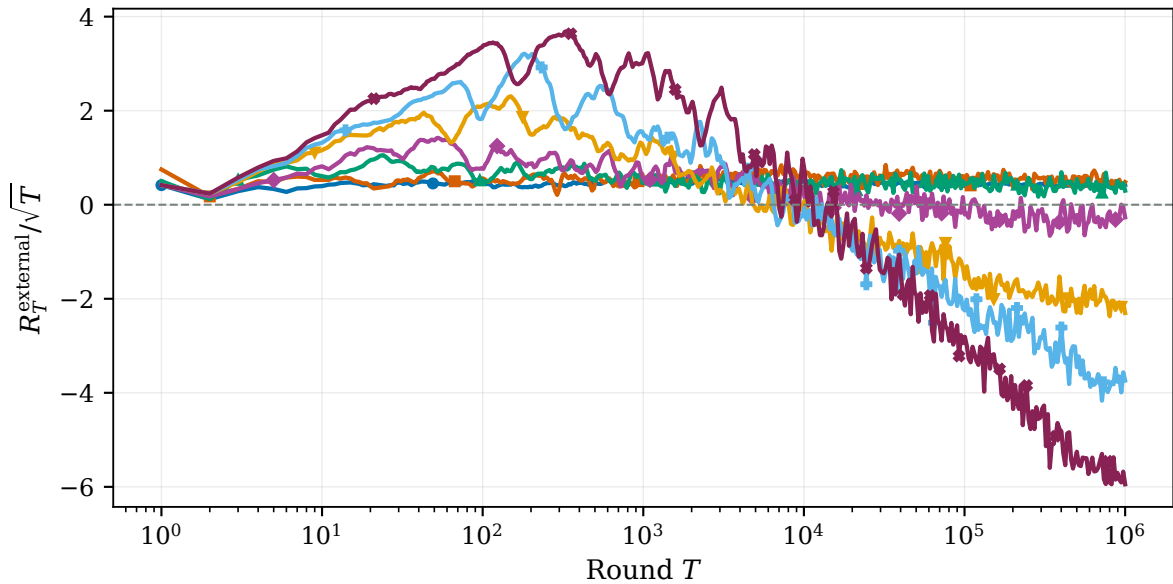
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Full information
Algorithm: SRM
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: External regret
View: R / T
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.000422167, SD = 0.000220374, SE = 6.36164e-05
K=3: mean = 0.000482583, SD = 0.000381884, SE = 0.00011024
K=5: mean = 0.000299333, SD = 0.0006334, SE = 0.000182847
K=10: mean = -0.000268167, SD = 0.000794326, SE = 0.000229302
K=20: mean = -0.00230258, SD = 0.000670909, SE = 0.000193675
K=30: mean = -0.00373942, SD = 0.000798321, SE = 0.000230455
K=50: mean = -0.00594458, SD = 0.00114787, SE = 0.000331362



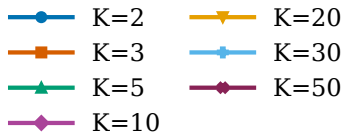
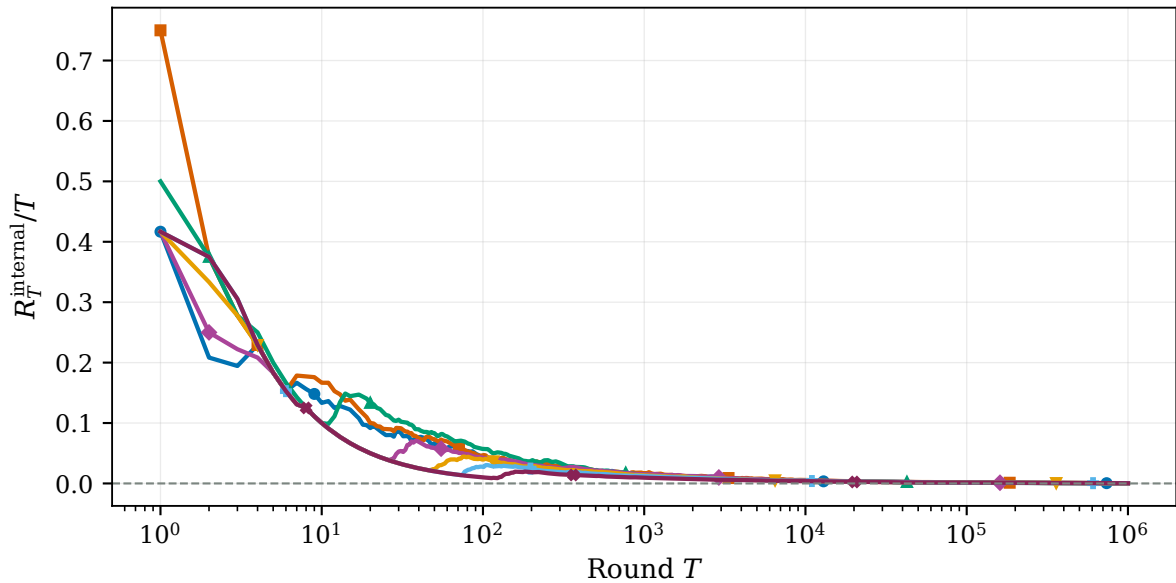
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Full information
Algorithm: SRM
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: External regret
View: $R / \sqrt{T}$
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.422167, SD = 0.220374, SE = 0.0636164
K=3: mean = 0.482583, SD = 0.381884, SE = 0.11024
K=5: mean = 0.299333, SD = 0.6334, SE = 0.182847
K=10: mean = -0.268167, SD = 0.794326, SE = 0.229302
K=20: mean = -2.30258, SD = 0.670909, SE = 0.193675
K=30: mean = -3.73942, SD = 0.798321, SE = 0.230455
K=50: mean = -5.94458, SD = 1.14787, SE = 0.331362



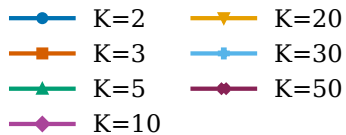
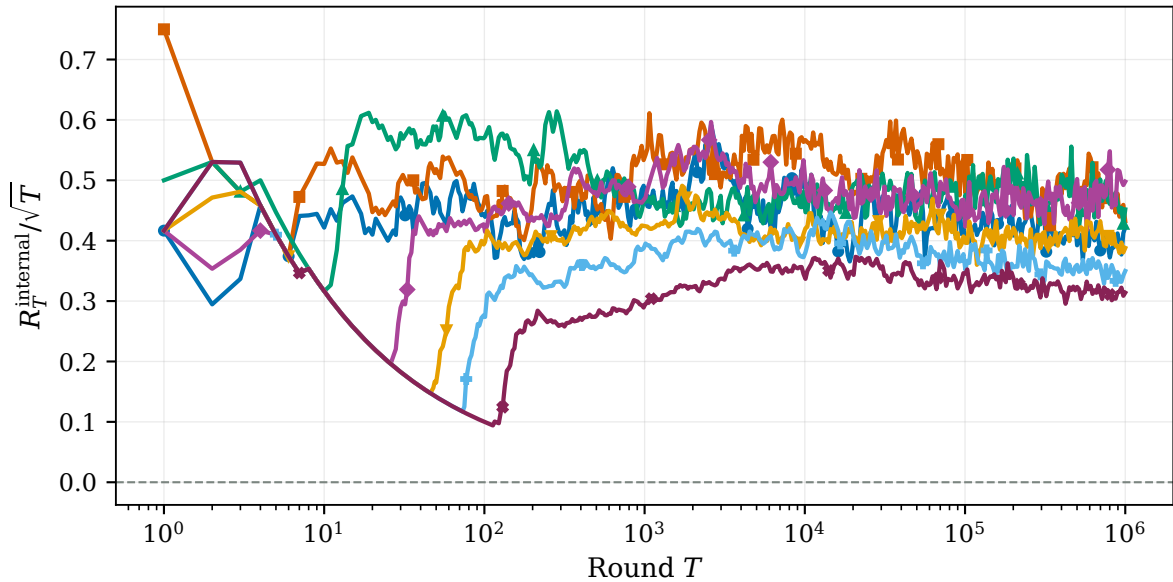
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Full information
Algorithm: SRM
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Internal regret
View: R / T
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.000422167, SD = 0.000220374, SE = 6.36164e-05
K=3: mean = 0.000432333, SD = 0.00024759, SE = 7.14731e-05
K=5: mean = 0.00045175, SD = 0.000132472, SE = 3.82413e-05
K=10: mean = 0.000497833, SD = 8.72695e-05, SE = 2.51925e-05
K=20: mean = 0.000388, SD = 6.4515e-05, SE = 1.86239e-05
K=30: mean = 0.000349833, SD = 4.33209e-05, SE = 1.25057e-05
K=50: mean = 0.000313917, SD = 2.77668e-05, SE = 8.01557e-06



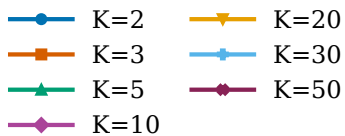
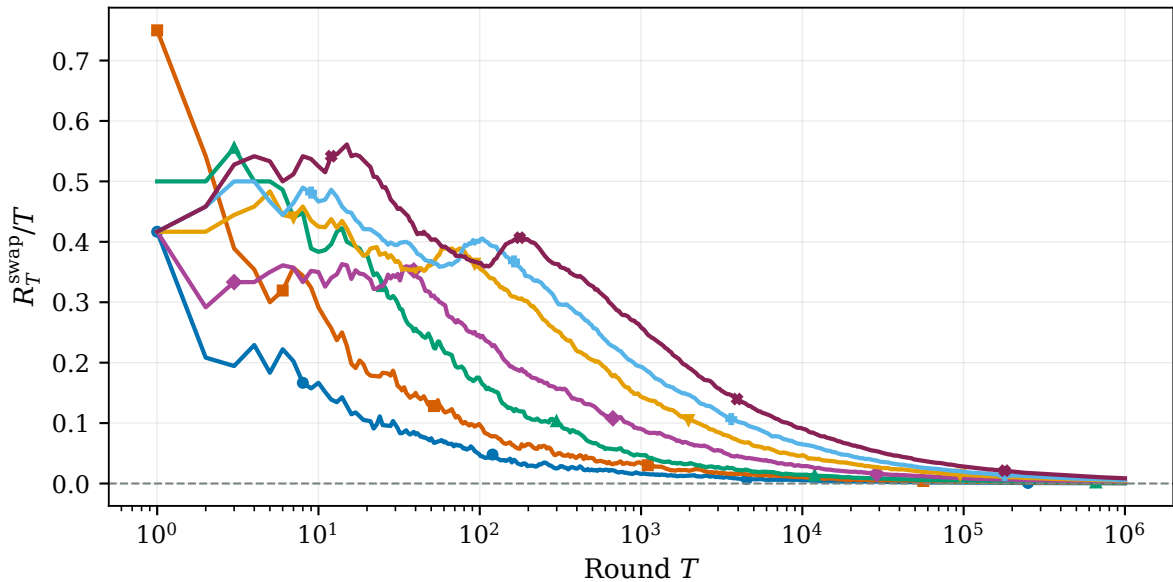
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Full information
Algorithm: SRM
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Internal regret
View: $R / \sqrt{T}$
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.422167, SD = 0.220374, SE = 0.0636164
K=3: mean = 0.432333, SD = 0.24759, SE = 0.0714731
K=5: mean = 0.45175, SD = 0.132472, SE = 0.0382413
K=10: mean = 0.497833, SD = 0.0872695, SE = 0.0251925
K=20: mean = 0.388, SD = 0.064515, SE = 0.0186239
K=30: mean = 0.349833, SD = 0.0433209, SE = 0.0125057
K=50: mean = 0.313917, SD = 0.0277668, SE = 0.00801557



Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Full information
Algorithm: SRM
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Swap regret
View: R / T
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.000459, SD = 0.000216719, SE = 6.25612e-05
K=3: mean = 0.00077025, SD = 0.000430279, SE = 0.000124211
K=5: mean = 0.00141558, SD = 0.000460341, SE = 0.000132889
K=10: mean = 0.00301133, SD = 0.000492072, SE = 0.000142049
K=20: mean = 0.00461258, SD = 0.000626724, SE = 0.00018092
K=30: mean = 0.00603983, SD = 0.000420336, SE = 0.000121341
K=50: mean = 0.00865742, SD = 0.000617749, SE = 0.000178329



Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Full information
Algorithm: SRM
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Swap regret
View: $R / \sqrt{T}$
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.459, SD = 0.216719, SE = 0.0625612
K=3: mean = 0.77025, SD = 0.430279, SE = 0.124211
K=5: mean = 1.41558, SD = 0.460341, SE = 0.132889
K=10: mean = 3.01133, SD = 0.492072, SE = 0.142049
K=20: mean = 4.61258, SD = 0.626724, SE = 0.18092
K=30: mean = 6.03983, SD = 0.420336, SE = 0.121341
K=50: mean = 8.65742, SD = 0.617749, SE = 0.178329

